MANGALORE UNIVERSITY

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"INVESTMENT PATTERN OF FOREIGN DIRECT INVESTMENT IN THE CORPORATE SECTOR"

A PROJECT REPORT SUBMITTED TO MANGALORE UNIVERSITY IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF MASTER OF BUSINESS ADMINISTRATION

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OCTOBER 2025

DECLARATION

I hereby declare that this project report titled **"INVESTMENT PATTERN OF FOREIGN DIRECT INVESTMENT(FDI) IN THE CORPORATE SECTOR"** has been prepared by me during the year 2024-2025, under the valuable guidance of **Dr. CHANDRASHEKHAR R,** Professor, Department of Business Administration, Mangalore University, in partial fulfillment of the requirements for the award of the Degree of Master of Business Administration.

I also declare that this project report is the result of my own efforts and it has not been submitted previously to any other University or Institution for the award of any degree or diploma

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This is to certify that this project report entitled

" INVESTMENT PATTERN OF FOREIGN DIRECT INVESTMENT(FDI) IN THE CORPORATE SECTOR"

is a bonafide work carried out by **NITHESH.K** (REGISTER NO: P05AZ23M015034) in partial fulfillment of the requirements for the award of the Degree of Master of Business Administration under my guidance and supervision.

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"Gratitude is not a think of expression: it is more a matter of feeling."

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Executive Summary

This project report examines the patterns and impact of Foreign Direct Investment (FDI) in India's corporate sector over the period 2015–2025. FDI has played a crucial role in accelerating industrial growth, improving competitiveness, and integrating Indian companies with global markets. Unlike portfolio investments, FDI involves long-term commitments such as equity participation, mergers, acquisitions, and wholly owned subsidiaries, making it a key driver of structural transformation in the economy.

The study was designed with clear objectives: to analyze trends and sectoral distribution of FDI inflows, evaluate modes of investment such as joint ventures and acquisitions, assess the role of FDI in enhancing competitiveness, and measure its contribution to capital formation, employment, and skill development. A purposive sampling approach was adopted, covering 64 companies across 16 sectors, including consumer goods, pharmaceuticals, automobiles, engineering, IT, construction, power, and tourism. Data spanning a ten-year period was analyzed using statistical tools like the two-sample t-test assuming unequal variances (Welch's t-test) and one-way ANOVA, enabling both pairwise and multi-sector comparisons of financial performance.

The study also underscores the broader implications of FDI. Policymakers can use these insights to refine sectoral policies and attract more strategic investment, while companies can benchmark themselves against peers to strengthen competitiveness. However, limitations such as dependence on secondary data, sector-specific focus, and exclusion of macroeconomic factors are acknowledged.

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CHAPTER 1
INTRODUCTION

INTRODUCTION:

Foreign Direct Investment (FDI) plays a central role in shaping the growth and competitiveness of the corporate sector. Unlike portfolio investments, FDI involves long-term commitments in the form of equity participation, joint ventures, mergers, acquisitions, and the establishment of wholly owned subsidiaries. For emerging economies like India, the corporate sector has been one of the primary recipients of FDI, as it provides both capital inflows and access to advanced technologies, managerial practices, and international markets.

The investment pattern in FDI within the corporate sector is influenced by several factors such as government policies, sectoral liberalization, market size, infrastructure, and global economic conditions. Over the past few decades, reforms in FDI regulations and the liberalization of caps in key industries have led to a steady inflow of foreign investments in sectors such as information technology, telecommunications, pharmaceuticals, manufacturing, and financial services. These inflows have supported modernization, enhanced productivity, and improved global integration of Indian companies.

Corporate FDI is not uniform across sectors. Capital-intensive industries such as infrastructure, energy, and automobiles often witness large equity stakes and joint ventures, while service-oriented sectors attract strategic alliances and technology partnerships. Similarly, investment routes vary: some corporations prefer the automatic route for ease of entry, while others rely on the government approval route in sensitive areas.

The evolving pattern of FDI also reflects global economic shifts. Multinational corporations seek to diversify supply chains, capture growing consumer demand in developing economies, and align with digital transformation. In return, host countries like India benefit through job creation, skill development, and higher tax revenues. Thus, the investment pattern of FDI in the corporate sector is not only a measure of financial inflows but also an indicator of long-term structural change in the economy.

OBJECTIVES OF THE STUDY:

1. To analyse the trends and patterns of FDI inflows into the corporate sector, with a focus on sector-wise distribution such as manufacturing, IT, telecommunications, pharmaceuticals, and financial services.

2. To examine the modes of foreign investment (greenfield investments, mergers and acquisitions, joint ventures, and wholly owned subsidiaries) commonly adopted by multinational corporations.
3. To understand the role of FDI in enhancing competitiveness of Indian companies through technology transfer, management expertise, and access to global markets.
4. To evaluate the contribution of FDI to economic growth in terms of capital formation, employment generation, and skill development within the corporate sector.
5. To explore emerging trends in FDI, particularly in new-age industries such as e-commerce, fintech, renewable energy, and digital services.
6. To test whether significant differences exist in company ratios across selected sectors.
7. To identify sectoral leaders and laggards in performance.
8. To provide analytical insights for investors, managers, and policymakers.

Hypothesis:

Introduction

In research and data analysis, it is often necessary to compare the means of multiple groups to determine if differences are meaningful or just due to random variation. The **Analysis of Variance (ANOVA)** is one of the most widely used statistical tools for this purpose. When there is only one independent variable, it is called a **single-factor or one-way ANOVA**. The central idea is to test whether all population means are equal or whether at least one group mean differs. This is achieved by formulating and testing hypotheses about population means.

Hypothesis in Single-Factor ANOVA

Analysis of Variance (ANOVA) is a statistical method used to compare the means of three or more groups to determine whether there are significant differences among them. In a **single-factor ANOVA**, also called a **one-way ANOVA**, there is only one independent variable (factor) with two or more levels, and the dependent variable is continuous. The goal is to test whether the variation in the dependent variable can be explained by differences between group means rather than by random chance.

Formulation of Hypotheses

In ANOVA, the hypotheses are set up in terms of population means. Let's assume there are *k* groups being compared, and the group means are represented as:

$\mu_1,\mu_2,\mu_3,\ldots,\mu_k$ \mu_1, \mu_2, \mu_3, …, \mu_k

where each μ represents the true population mean of a group.

- **Null Hypothesis (H_0):**

All group means are equal.
Mathematically,

$H_0:\mu_1=\mu_2=\mu_3=\ldots=\mu kH_0$: \mu_1 = \mu_2 = \mu_3 = … = \mu_k

This implies that the factor (independent variable) has no effect on the dependent variable. Any observed difference among sample means is due to random variation or sampling error.

- **Alternative Hypothesis (H_1):**

At least one group mean is different from the others.
Mathematically,

H_1:At least one $\mu i\neq\mu j$ for some $i\neq jH_1$: \text{At least one } \mu_i \ne_q \mu_j \quad \text{for some } i \neq_j

This does not specify which groups differ or how many are different; it only states that not all means are equal.

Testing the Hypotheses

The test is carried out by partitioning the total variability in the data into two components:

1. **Between-group variability**: Variation due to differences among the group means.

2. **Within-group variability:** Variation within each group, often referred to as error or residual variation.

The **F-statistic** is then computed as:

F=Mean Square Between Mean Square Within F = \frac{\text{Mean Square Between}}{\text{Mean Square Within}}

- A high F-value indicates that between-group variability is large relative to within-group variability, suggesting that group means are not equal.

- A low F-value suggests that any differences between group means are small relative to within-group variability, so the null hypothesis may hold.

Decision Rule

- Compare the calculated F-value with the critical value from the F-distribution table at a chosen significance level (e.g., $\alpha = 0.05$).
- Alternatively, compare the **p-value** with α:
  - If $\mathbf{p \leq \alpha}$, reject H_0 and conclude that there is a significant difference among the group means.
  - If $\mathbf{p > \alpha}$, fail to reject H_0, meaning there is insufficient evidence to claim group means are different.

t-Test: Two-Sample Assuming Unequal Variances

Introduction

The **t-Test: Two-Sample Assuming Unequal Variances**, also known as **Welch's t-test**, is a statistical method used to determine whether there is a significant difference between the means of two independent groups when their population variances are not assumed to be equal. This is an extension of the independent samples t-test, adjusted to handle cases where the assumption of equal variances is violated.

This test is widely used in business, economics, and social sciences research, where real-world data often show unequal variability between groups.

Purpose of the Test

- To compare the means of two unrelated groups.
- To test if any observed difference is statistically significant rather than due to sampling error.
- To provide evidence in decision-making, such as evaluating the performance of two companies, comparing market strategies, or testing customer responses across two groups.

Hypotheses

In hypothesis testing, we set up two opposing statements:

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2$

- The two population means are equal. There is no significant difference.

- **Alternative Hypothesis (H_1):** $\mu_1 \neq \mu_2$ (two-tailed test)
  - The two population means are not equal. There is a significant difference.

Depending on the research question, the alternative hypothesis may also be one-tailed ($\mu_1 > \mu_2$ or $\mu_1 < \mu_2$).

Formula

The test statistic (t) is calculated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{X}_1, \bar{X}_2$= sample means
- s_1^2, s_2^2= sample variances
- n_1, n_2= sample sizes

Since the two samples are assumed to have unequal variances, the **degrees of freedom (df)** are calculated using the **Welch–Satterthwaite equation**:

$$df = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

This df is often a non-integer value.

Decision Rule

1. Choose a significance level (α), commonly 0.05.
2. Calculate the t-statistic using the formula.
3. Determine the degrees of freedom.
4. Compare the calculated t-statistic to the critical t-value from the t-distribution table.
   - If $|t| >$ t-critical → Reject H_0 (significant difference).
   - If $|t| \leq$ t-critical → Fail to reject H_0 (no significant difference).
5. Alternatively, compare the p-value to α:
   - If $p \leq \alpha$ → Reject H_0.
   - If $p > \alpha$ → Fail to reject H_0.

Assumptions of the Test

1. The two groups are independent of each other.
2. The dependent variable is continuous (measured on interval/ratio scale).
3. The data in each group are approximately normally distributed.
4. The variances of the two groups are **unequal** (or unknown), making this test more appropriate than the equal variance version.

Sample of the study:

The study follows a purposive sampling approach, where companies were selected based on relevance, availability of data, and representation of key sectors in the Indian economy. Since the objective of the research was to compare financial performance across industries, the sample was chosen to cover a diverse set of sectors rather than focusing on a single industry.

Selection Criteria:

1. **Availability of Data:** Companies with at least ten years of consistent and reliable financial information (2015–2025) were included.
2. **Industry Representation:** At least two to five companies were chosen from each major sector (where possible) to allow for meaningful intra-sector and inter-sector comparisons.
3. **Market Significance:** Preference was given to companies that hold a notable presence in their sector, either in terms of market share, reputation, or operational scale.
4. **Comparability:** Companies were selected where the primary performance measure—percentage of capital per share—was consistently reported, allowing uniform analysis.

Sampling Frame:

The final sample consists of over 60 companies across 16 industries, ranging from traditional sectors like engineering, construction, and chemicals to high-growth sectors such as pharmaceuticals, consumer goods, power, and IT services. This broad frame ensures that the analysis reflects both established and emerging industries.

Sampling Technique:

- For industries with many players (e.g., FMCG, automobiles, pharmaceuticals), only a subset of leading or representative companies was selected.
- For niche or specialized sectors (e.g., hotel & tourism, machine tools), only one or two companies were available, and these were included in full.
- In some cases, companies with consistent but identical results (e.g., Kennametal India Ltd. and John Cockerill India Ltd. in machine tools) were retained to highlight statistical issues such as zero variance.

Rationale for Period (2015–2025):

The ten-year time frame provides a long enough horizon to observe trends, stability, and fluctuations in performance. It also allows the use of robust statistical techniques such as ANOVA and t-tests, which require multiple observations to generate meaningful results.

Strength of the Sample:

The chosen sample is large, diverse, and balanced across industries. It provides a strong foundation for statistical analysis and ensures that results are not biased toward one particular sector. This makes the findings relevant for understanding financial performance patterns across the broader corporate landscape in India.

SOURCES OF DATA:

The present study is based on **secondary data** collected from official and reliable sources. Information on Investment pattern of FDI in the corporate sector was primarily obtained from the NSDL report, which provides official records of corporate events. To ensure accuracy and completeness, additional verification was performed using **screener.in** databases.

The study considers a filtered sample of **64 companies**, selected from companies that announced yearly during the study period. Yearly closing prices of these companies were collected for a period of **10 years after**, to facilitate event study analysis.

Additional corporate data, including no. of share price & percentage of capital, were collected; however, the analysis in this study focuses primarily to calculate ANOVA: single factor, t-Test: Two-Sample Assuming Unequal Variances and graphical presentation.

Scope of the Study

The study on investment patterns in Foreign Direct Investment (FDI) for the corporate sector aims to provide a comprehensive analysis of how foreign capital flows into corporate enterprises over time. The scope is defined both in terms of the parameters of data analysis and the practical implications for policy makers and corporate stakeholders.

1. **Time Frame:**
   The study covers a 10-year period, from 2015-16 to 2024-25. This allows for the identification of long-term trends, cyclical patterns, and shifts in investment preferences across different economic and policy environments.

2. **Geographical Coverage:**
   The focus is on FDI inflows into companies operating in the chosen country (or countries, if comparative), highlighting the sectors most attractive to foreign investors. Regional variations, if any, may also be analyzed to understand localized investment trends.

3. **Sectoral Analysis:**
   The study includes 64 companies across 16 sectors. This allows for cross-sectoral comparisons to identify which industries attract more FDI, the growth trajectory of investment, and the sector-specific factors influencing investor decisions.

4. **Investment Type:**
   The research primarily focuses on corporate FDI, which includes equity investment, reinvested earnings, and intra-company loans. Other forms of foreign investment, such as portfolio investment, are not considered, keeping the study targeted on long-term and strategic investment behavior.

5. **Data and Methodology:**
   Quantitative analysis is the core approach, employing statistical tools such as ANOVA (Single Factor), t- Test: Two-Sample Assuming Unequal Variances to identify differences in investment patterns among companies and sectors. Graphical representations will supplement statistical findings to provide clear visual insights.

6. **Policy and Strategic Insights:**
   The study can guide policymakers in understanding which sectors are

attracting FDI and why. It also informs corporate strategy by revealing the dynamics of foreign investment, helping companies align their practices to attract or retain FDI.

7. **Limitations Within Scope:**
   While the study examines investment patterns, it does not deeply analyze investor motivations, political risk factors, or macroeconomic conditions beyond their immediate impact on FDI inflows. Similarly, only historical data is analysed; predictive modeling or forecasts are not the main focus.

Methodology of the Study:

I. t-Test: Two-Sample Assuming Unequal Variances

The two-sample t-test assuming unequal variances (also called Welch's t-test) is used to compare the means of two independent groups when their variances are not assumed to be equal. In this study, the test was applied to evaluate whether there is a statistically significant difference between the performance measures (such as profitability ratios, revenue growth, or any other financial metric considered) of two selected companies.

Steps followed:

1. **Formulation of Hypotheses:**
   - **Null Hypothesis (H_0):** There is no significant difference between the mean values of the two companies.
   - **Alternative Hypothesis (H_1):** There is a significant difference between the mean values of the two companies.
2. **Selection of Test:** The *Two-Sample t-Test Assuming Unequal Variances* was chosen because the companies may have different financial structures and risk exposures, leading to unequal variances in their performance measures.
3. **Calculation of Test Statistic:** The formula compares the difference in sample means relative to the variability of both groups. The test statistic is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- (\bar{X}_1, \bar{X}_2) = sample means,
- (s_1^2, s_2^2) = sample variances,
- (n_1, n_2) = sample sizes.

4. **Decision Rule:** The calculated t-value and the corresponding p-value were compared with the significance level (commonly 5%). If $p < 0.05$, the null hypothesis was rejected, indicating a significant difference in performance between the two companies.

II. ANOVA: Single Factor

Analysis of Variance (ANOVA) is a statistical technique used to test whether there are significant differences between the means of more than two groups. In this study, the *Single-Factor ANOVA* was applied to compare the performance of multiple companies over a defined period.

Steps followed:

1. **Formulation of Hypotheses:**
   - **Null Hypothesis (H_0):** All company means are equal (no significant difference in performance).
   - **Alternative Hypothesis (H_1):** At least one company's mean is significantly different.
2. **Rationale for Test:** ANOVA was chosen because it allows simultaneous comparison of multiple groups, avoiding the need for multiple pairwise t-tests which could increase the risk of error.
3. **Calculation of Test Statistic:**
   ANOVA partitions the total variation in the data into two components:
   - **Between-group variation:** Differences due to variation between company means.
   - **Within-group variation:** Differences due to random fluctuations within each company's performance.

The F-statistic is calculated as:

[F = \frac{\text{Mean Square Between Groups (MSB)}}{\text{Mean Square Within Groups (MSW)}}]

- MSB = Between-group sum of squares / (k – 1)

- MSW = Within-group sum of squares / (N – k)
- where k = number of groups, N = total observations.

4. **Decision Rule:** The calculated F-value was compared with the critical F-value from statistical tables at the chosen significance level (usually 5%). If the calculated F exceeded the critical F, or if the p-value was less than 0.05, the null hypothesis was rejected.

Application in the Study

- The **t-test** was applied when comparing two companies to test for differences in their average performance.
- The **ANOVA** test was applied when more than two companies were compared simultaneously to determine whether at least one company showed a statistically different mean performance compared to others.

Limitation of the Study:

1. **Data Availability** – Acknowledge gaps and rely on multiple sources (government reports, international databases, and corporate disclosures) to improve reliability.
2. **Time Frame Restriction** – Clearly define why the chosen period is relevant (e.g., post-policy reform, pre- and post-crisis) and suggest future studies covering longer horizons.
3. **Geographical Scope** – State that findings are context-specific and propose comparative studies across countries to test external validity.
4. **Sectoral Focus** – Justify the focus on the corporate sector as a significant driver of FDI, while recommending that future research explore cross-sectoral comparisons.
5. **Macroeconomic Influences** – Control for key macroeconomic indicators in the analysis and clarify that external shocks are beyond the scope but can be explored in extended models.

6. **Policy Changes** – Highlight that findings reflect the policy environment during the study period, and recommend updating analyses periodically to capture new reforms.
7. **Firm-Specific Factors** – Acknowledge that qualitative aspects (like firm strategy or leadership) are difficult to quantify, and suggest case studies or interviews as complementary methods.
8. **Measurement Issues** – Define "investment patterns" explicitly (e.g., type of FDI, entry mode, capital intensity) and use standardized classifications to minimize ambiguity.
9. **Causality vs. Correlation** – Clarify that the study is exploratory/descriptive, not predictive, and suggest advanced econometric methods (like panel regressions or instrumental variables) for future causal inference.
10. **Generalizability of Findings** – Emphasize that the study provides insights into a specific sample and time, and encourage replication with broader datasets to strengthen general conclusions.

MODULE-2

FDI COMPANY PROFILE AND CONCEPTUAL FRAMEWORK

FDI COMPANIES PROFILE:

I. MISCELLANEOUS MECHANICAL & ENGINEERING:

1) **De Nora India Ltd.:**
   De Nora India Ltd. is a subsidiary of Oronzio De Nora International B.V., based in Italy. The company operates in the field of electrochemical processes and provides sustainable technologies and solutions for water treatment and energy storage. With strong backing from its Italian parent company, De Nora India plays a significant role in serving both industrial and environmental sectors across the region.
2) **Disa India Ltd.:**
   Disa India Ltd. is supported by two prominent European companies—DISA Holding AG from Switzerland and DISA Holding A/S from Denmark. Disa India Ltd. specializes in manufacturing and supplying foundry machinery and surface preparation equipment. Its parent companies are globally recognized leaders in foundry technology, contributing to the company's robust engineering capabilities and innovative product offerings in India and beyond.

II. CONSUMER GOODS

1) **Hindustan Unilever Ltd.:**
   Hindustan Unilever Ltd. is a leading fast-moving consumer goods (FMCG) company in India and a subsidiary of **Unilever PLC**, headquartered in the United Kingdom. It offers a wide range of products in the areas of personal care, home care, foods, and beverages, and is one of the most trusted household names in India.
2) **Colgate- Palmolive (India) Ltd.:**
   Colgate-Palmolive (India) Ltd. is part of the global **Colgate-Palmolive Co.**, based in the United States, with significant support from **Colgate-Palmolive (Asia) Pte. Ltd**, Singapore, and **Norwood International Incorporated**, United States. The company is a market leader in oral care and also has a presence in personal and home care segments.
3) **Gillette (India) Ltd.:**
   Gillette (India) Ltd. operates under **Procter and Gamble India Holdings B.V.**, based in the Netherlands. Known primarily for its shaving products, Gillette has a strong brand presence and is a key player in the Indian grooming market.
4) **Nestle India Ltd.:**

Nestlé India Ltd. is a subsidiary of **Nestlé S.A.**, Switzerland—one of the world's largest food and beverage companies. Nestlé India manufactures and markets a variety of products in nutrition, health, and wellness, with well-known brands like MAGGI under **MAGGI Enterprises Limited**, also based in Switzerland.

5) **Bata India Ltd**.

Bata India Ltd., incorporated in 1931, is one of India's largest footwear manufacturers and retailers. Headquartered in Gurugram, the company operates a wide network of retail stores and also supplies institutional and industrial footwear. It is part of the global **Bata Shoe Organization**, with its parent entity being **Bata (BN) BV, Amsterdam, Netherlands**. Bata India combines international design expertise with local market insights, offering a diverse portfolio of shoes, accessories, and lifestyle products that cater to different consumer segments across the country.

III. SERVICE SECTOR, TRADING

1) **Infinite computer solutions (India) Ltd.:**

Infinite Computer Solutions (India) Ltd. is a global technology company providing IT services, platforms, and solutions. It operates under the umbrella of **Infinite Technologies, LLC**, based in the United States. The company specializes in infrastructure management, mobility, and product engineering, serving clients across various industries.

2) **Flipkart:**

Flipkart is one of India's largest e-commerce platforms and is currently owned by **Walmart**, a global retail giant headquartered in the United States. With Walmart's backing, Flipkart continues to expand its footprint across India, offering a wide range of products from electronics to fashion and groceries.

3) **Essar Shipping Ltd:**

Essar Shipping Ltd. is part of the larger Essar Group and is engaged in transportation of crude oil, petroleum products, and dry bulk cargo. The company is associated with **Essar Port & Shipping Limited** in Mauritius and **Essar Shipping & Logistics Limited** in Cyprus, enhancing its international logistics and maritime capabilities.

IV. MI, GLASS, CERMAICS & GYPSUM

1) **Saint- Gobain- Sekurit India Ltd.**

Incorporated in 1973, Saint-Gobain Sekurit India Ltd. specializes in manufacturing automotive safety glass, including laminated and tempered glass products. Operating from its facility in Chakan, Pune, the company caters to original equipment manufacturers (OEMs), transport sectors, and the aftermarket. As part of the global Saint-Gobain Group, it benefits from over 350 years of expertise in materials and glazing solutions.

2) **Esab India Ltd**.

Established in 1987, ESAB India Ltd. is a leading provider of welding and cutting equipment, along with consumables. The company offers a comprehensive range of solutions for various applications, serving industries both within India and internationally. ESAB India is a subsidiary of ESAB Holdings Limited, based in the UK.

3) **Foseco India Ltd.**

Founded in 1958, Foseco India Ltd. manufactures products for the metallurgical industry, including additives and consumables that enhance the physical properties and performance of castings. The company operates in the steel and foundry sectors, providing solutions such as binders, coatings, and refractory materials. In August 2025, Foseco announced an agreement to acquire a 75% stake in Morganite Crucible (India) Ltd. for ₹653 crore, marking a significant expansion in its portfolio.

4) **Morganite Crucible (India) Ltd.**

Morganite Crucible (India) Ltd., founded in 1986 in Aurangabad, Maharashtra, manufactures high-performance crucibles and foundry consumables for ferrous and non-ferrous metal industries. Originally promoted by Greaves Cotton & Company and Morganite Crucible Limited, UK, it is now part of Morgan Advanced Materials' Molten Metal Systems division. In August 2025, UK-based Foseco acquired a 75% stake in the company for ₹653 crore to strengthen its industrial goods and engineering capabilities. **Morganite Crucible Limited (UK)**, incorporated in 1987 in Windsor, Berkshire, is a private, non-trading company. **Morgan Terrassen B.V. (Netherlands)** is a holding company with a 36.5% stake in Morganite Crucible (India) Ltd., reflecting its strategic interest in the Indian subsidiary.

5) **Vesuvius India Ltd.**

Incorporated in 1991, Vesuvius India Ltd. is engaged in the manufacturing, trading, and sale of refractory products used in the iron and steel industries. The company's offerings include monoblock stoppers, ladle shrouds, tundish nozzles, and other flow control solutions. Vesuvius India is part of the global

Vesuvius Group, which provides engineering services and solutions to the metal flow industry.

V. INDUSTRIAL INSTRUMENTS, PRIME MOVER

1) **Integra Engineering India Ltd.**
   Founded in the 1980s and listed on the Bombay Stock Exchange, Integra Engineering India Ltd. is a leading supplier of railway and rolling stock components. Headquartered in India, the company designs and manufactures propulsion system components, metro and railway interiors, control systems, and enclosures. It is a subsidiary of Integra Holding AG, a Swiss management holding company with global investments.
2) **Singer India Ltd.**
   Established in 1977, Singer India Ltd. is a prominent manufacturer of sewing machines and home appliances. The company operates as a subsidiary of Singer B.V., Netherlands, which holds a 51% stake. Previously known as Singer India B.V., it has undergone significant changes, including the sale of a 42.4% equity stake by Retail Holding Asia B.V. to an international private investor in December 2020.
3) **Stovec industries Ltd.**
   Incorporated in 1973, Stovec Industries Ltd. is a subsidiary of SPGPrints B.V., Netherlands. The company specializes in providing holistic printing solutions, including manufacturing machines and parts for textile printing, graphic printing, and sugar screens. Stovec operates as a competence center for development, production, installation, after-sales service, and support.
4) **Ingersoll- Rand (India) Ltd.**
   Ingersoll Rand (India) Ltd. is a subsidiary of Ingersoll Rand Inc., a U.S.-based industrial company. Ingersoll Rand Inc. specializes in manufacturing and distributing compressed air solutions, power tools, lifting equipment, and fluid management systems. The company operates globally, with a strong presence in India, offering a wide range of products and services to various industries.

VI. ELECTRICAL EQUIPMENT, Electronics

1) **Indo Tech Transformers Ltd.**
   Established in 1977, Indo Tech Transformers Ltd. is a leading manufacturer of power and distribution transformers in India. The company specializes in producing high-quality transformers for various applications, including industrial, commercial, and utility sectors. In 2009, Mexico-based Prolec GE

Internacional, S.DE R.L.DE.C.V acquired a majority stake in Indo Tech Transformers, enhancing its global presence in the transformer manufacturing industry.

2) **Panasonic carbon India Co. Ltd**.

Panasonic Carbon India Co. Ltd., a subsidiary of Panasonic Corporation, is the sole manufacturer of high-standard carbon rods in India. The company plays a crucial role in the battery industry by supplying carbon rods used in dry cell batteries. Recently, Panasonic has been restructuring its operations in India, exiting the refrigerator and washing machine segments to focus on more profitable areas such as HVAC and B2B solutions.

3) **Honeywell Automation India Ltd**.

Honeywell Automation India Ltd. (HAIL), a subsidiary of Honeywell Asia Pacific Inc., is a leading provider of integrated automation and software solutions in India. Incorporated in 1984, HAIL offers a wide range of products and services, including process solutions and building solutions. The company is set to surpass $1 billion in revenue in 2025, driven by advancements in AI, 5G, and cloud technologies.

4) Siemens Ltd.

Siemens Ltd., a subsidiary of Siemens International Holding B.V., is a prominent technology company in India. The company specializes in industrial automation, building automation, and healthcare solutions. Siemens Metals Technologies Vermögensverwaltungs GmbH, an Austrian entity, holds a stake in Siemens Ltd., reflecting the company's global operations and investments.

5) Ricoh India Ltd.

Ricoh India Ltd., formerly a subsidiary of Ricoh Company Limited, was a leading provider of imaging and printing solutions in India. In 2021, Ricoh concluded the transfer of all its shares in Ricoh India Ltd. to third parties, dissolving its capital relationship with the company.

6) D- Link (India) Ltd.

D-Link (India) Ltd., a subsidiary of D-Link Holding Mauritius Inc., is a leading provider of networking and communication solutions in India. The company offers a wide range of products, including routers, switches, and wireless solutions, catering to both consumer and enterprise markets.

7) Whirlpool of India Ltd.

Whirlpool of India Ltd., a subsidiary of Whirlpool Mauritius Ltd., is a major player in the home appliance industry in India. In 2024, Whirlpool sold a 24% stake in its Indian subsidiary for approximately $468 million, reducing its ownership from 75% to 51%.

8) **Timex Group India Ltd.**

Timex Group India Ltd., a subsidiary of Timex Group Luxury Watches BV, is a prominent watch manufacturer in India. The company offers a wide range of timepieces under various brands, catering to different market segments.

VII. DRUGS & PHARMACEUTICALS

1) **Abbott India Limited**

Abbott India Limited, a leading healthcare and pharmaceutical company in India, is part of the global Abbott group headquartered in the U.K. Its Indian operations are supported by subsidiaries including Abbott Capital India Limited, Abbott Healthcare Products Limited, and British Colloids Limited. The company produces a wide range of pharmaceuticals, nutritional products, diagnostics, and medical devices.

2) **Astrazeneca Pharma India Ltd.**

AstraZeneca Pharma India Ltd. is a subsidiary of AstraZeneca Pharmaceuticals AB, Sweden. The company focuses on prescription medicines, primarily in oncology, cardiovascular, renal, and respiratory therapies, with a strong presence in the Indian pharmaceutical market.

3) **Sanofi India Ltd.**

Sanofi India Ltd., part of Germany-based Hoechst GmbH, develops and markets pharmaceuticals, vaccines, and consumer healthcare products in India. The company emphasizes chronic disease management, vaccines, and innovative medicines.

4) **Galxo Smithkline Pharmaceuticals Ltd**.

GlaxoSmithKline Pharmaceuticals Ltd. (GSK) in India is part of the UK-based Glaxo Group Limited. Other related entities include GlaxoSmithKline Pte Limited (Singapore), Eskaylab Limited (U.K.), Burroughs Wellcome International Limited (U.K.), and Castleton Investment Limited (Mauritius). GSK India produces a wide range of pharmaceuticals, vaccines, and consumer healthcare products.

5) **Novartis India Ltd.**

Novartis India Ltd. is a subsidiary of Novartis AG, Switzerland. The company develops and markets innovative pharmaceuticals and generics across multiple therapeutic areas, including oncology, ophthalmology, and cardiovascular diseases.

6) **Pfizer Ltd.**

Pfizer Ltd. operates in India under Pfizer East India B.V. (Netherlands) and is linked to several U.S.-based entities, including Wyeth LLC, Wyeth Holdings

Corporation, Warner-Lambert Company LLC, Parke, Davis & Company LLC, Pharmacia Corporation, and John Wyeth & Brother Ltd (U.K.). Pfizer focuses on vaccines, oncology, cardiology, and various prescription medicines in the Indian market.

VIII. CHEMICALS (OTHER THAN FERTILIZERS)

1) **BASF India Ltd.**
   BASF India Ltd. is the Indian subsidiary of BASF Societas Europaea, Germany. It also has ties to BASF Construction Solutions GmbH (Germany) and BASF Schweiz AG (Switzerland). The company operates in chemicals, plastics, performance products, agricultural solutions, and coatings, serving multiple industrial sectors.
2) **Sunshield Chemicals Ltd.**
   Sunshield Chemicals Ltd., based in India, is a subsidiary of Rhodia Amines Chemicals Pte Ltd., Singapore. The company specializes in specialty chemicals for coatings, pharmaceuticals, and industrial applications.
3) **Elantas Beck (India) Ltd.**
   Elantas Beck (India) Ltd. operates under ELANTAS GmbH, Germany (Wesel). The company manufactures insulating materials, varnishes, and coatings for the electrical and industrial sectors.
4) **Berger paints India Ltd.**
   Berger Paints India Ltd., a leading paint manufacturer in India, is linked to Jenson Nicholson (Asia) Limited, U.K. The company produces decorative paints, industrial coatings, and protective coatings for multiple sectors.
5) **Dic India Ltd**.
   Dic India Ltd. is part of Dic Asia Pacific Pte. Ltd., Singapore. It specializes in specialty chemicals, pigments, and industrial coatings for a variety of applications.
6) **3M India Limited**
   3M India Limited is a subsidiary of 3M Company, USA. It operates in healthcare, consumer goods, electronics, and industrial products, providing innovative solutions across multiple sectors.
7) **Akzo Nobel India Limited**
   Akzo Nobel India Limited is the Indian arm of Akzo Nobel NV, Netherlands. The company manufactures paints, coatings, and specialty chemicals for decorative, industrial, and protective applications.

IX. Construction (Infrastructure)

1) **ITD Cementation India Ltd**.

ITD Cementation India Ltd. is a leading construction and civil engineering company in India, specializing in infrastructure projects such as bridges, roads, ports, and buildings. It is a subsidiary of **Italian-Thai Development Public Company Limited**, headquartered in Bangkok, Thailand, which provides strategic guidance, technical expertise, and financial support for large-scale construction projects. The company combines international experience with local execution capabilities to deliver complex engineering projects across India.

2) Johnson Controls-Hitachi Air Condition. India Ltd

Johnson Controls-Hitachi Air Conditioning India Limited (JCHAI) is a key player in India's residential and light commercial air-conditioning market. Headquartered in Kadi, Gujarat, the company manufactures a wide range of air conditioners under the Hitachi Cooling & Heating brand and distributes them through a nationwide dealer and service network. Originally part of the Johnson Controls–Hitachi global joint venture formed in 2015, JCHAI became majority-owned by Robert Bosch GmbH in July 2025. The acquisition positions the company within Bosch's Home Comfort division, strengthening its ability to expand in India's growing HVAC sector with advanced technology and global expertise.

X. POWER

1) ABB India Limited

ABB India Limited is a subsidiary of **ABB Limited, Zurich**, a global leader in electrification and automation. The company operates across sectors such as power grids, robotics, motion, and industrial automation, offering advanced technologies for energy efficiency and digitalization in India.

2) G E Power India Ltd.

GE Power India Ltd. is linked to **Alstom Finance BV, Netherlands**, and is part of GE's global power division. The company provides technology, engineering, manufacturing, project management, and supply of products and services for power plants, including boilers, turbines, and environmental control systems.

3) Cummins India Ltd.

Cummins India Ltd. is a subsidiary of **Cummins Inc., USA**, a world leader in engine design, manufacturing, and related technologies. In India, the

company produces diesel and natural gas engines, power generation systems, and components for automotive, industrial, and power sectors.

4) **Honda Siel Power Products Ltd**.

Honda Siel Power Products Ltd., a subsidiary of **Honda Motor Co., Ltd., Japan**, manufactures and markets a wide range of power products, including portable generators, water pumps, tillers, and general-purpose engines. It serves both domestic and international markets with a focus on reliability and fuel efficiency.

XI. HOTEL & TOURISM

1) Thomas Cook (India) Ltd.

Thomas Cook (India) Ltd. is a leading integrated travel and travel-related financial services company in India. It offers a wide portfolio that includes foreign exchange, corporate travel, leisure travel, visa services, travel insurance, and MICE (Meetings, Incentives, Conferences, and Exhibitions). The company is owned by **Fairbridge Capital (Mauritius) Limited**, an investment arm of Fairfax Financial Holdings, which took control in 2012. With this backing, Thomas Cook India has expanded its presence across India and other international markets, strengthening its position as a trusted travel services provider.

XII. PETROLEUM& NATURAL GAS

1) **Castrol India Ltd**.

Castrol India Ltd. is one of the leading lubricant companies in the country, engaged in manufacturing and marketing automotive, industrial, and marine lubricants. It is a subsidiary of **Castrol Limited, London, United Kingdom**, which is part of BP Group. The company operates blending plants in India and has a strong distribution network serving both retail and industrial customers.

2) Gulf Oil Lubricants India Ltd

Gulf Oil Lubricants India Ltd., part of the Hinduja Group, manufactures and markets a wide range of automotive and industrial lubricants. Its parent entity is **Gulf Oil International (Mauritius) Inc.**, which manages the Gulf brand globally. The company has a strong market presence in engine oils, lubricants, and allied products across India.

XIII. AGRICULTURAL SERVICE, TEA & COFFEE

1) **Bayer cropscience Ltd.**

Bayer CropScience Ltd. is the Indian subsidiary of **Bayer AG, Germany**, with additional links to **Bayer CropScience AG, Germany**, and **Bayer SAS, France**. The company focuses on agricultural inputs, including crop protection products, seeds, and non-agricultural pest control. Following Bayer's global acquisition of **Monsanto Company (USA)** in 2018, Monsanto's portfolio (notably seeds and traits) has been integrated into Bayer's crop science business, strengthening its presence in India.

2) **Monsanto India Ltd.**

Monsanto India Ltd., formerly a subsidiary of **Monsanto Company, USA**, specialized in agricultural biotechnology, seeds (especially hybrid maize and cotton), and crop protection chemicals. After Monsanto's global acquisition by Bayer, its Indian operations became part of Bayer's CropScience business.

3) **Goodricke Group Ltd.**

Goodricke Group Ltd., incorporated in 1977, is a leading producer of tea in India. It operates multiple tea estates across West Bengal and Assam, manufacturing both CTC and orthodox teas. Its parent companies are **Western Dooars Investments Ltd., England**, and **Assam Dooars Investments Ltd., U.K.**, reflecting its historic ties to British investments in the Indian tea industry.

XIV. MACHINE TOOLS

1) **Kennametal India Ltd.**

Kennametal India Ltd. is a leading manufacturer of hard metal products, cutting tools, and engineered components, serving industries such as aerospace, automotive, mining, construction, and energy. The company operates from its state-of-the-art facility in Bengaluru, Karnataka, and supplies both to the domestic and export markets. It is backed by two parent entities: **Meturit AG, Switzerland**, and **Kennametal Inc., United States**, a global leader in materials science and tooling solutions. Through this support, Kennametal India benefits from advanced technology, global expertise, and a strong innovation pipeline.

2) **John Cockerill India Ltd**

John Cockerill India Limited is part of the global John Cockerill Group, a Belgium-based engineering and industrial solutions company with a long history in providing mechanical equipment, technology, and services. The Indian subsidiary was originally established to support the steel, energy, and environment sectors with specialized industrial equipment and integrated

solutions. Its ownership is linked to CMI Industry Automation Private Limited and Cockerill Maintenance & Ingénierie S.A., both headquartered in Belgium. In India, the company operates manufacturing facilities and engineering centers that cater to industries such as steel processing, heat recovery systems, and industrial automation. Through its integration with the global John Cockerill network, the company leverages advanced European technology while adapting solutions for the Indian market. With expertise in design, fabrication, installation, and servicing, John Cockerill India has positioned itself as a reliable partner in delivering sustainable and technologically advanced industrial solutions.

XV. PAPER & PLUP, PRINTING

1) **Voith Paper Fabrics India Ltd.**

Voith Paper Fabrics India Ltd. is a leading manufacturer of paper machine clothing, which includes forming, press, and dryer fabrics used in the paper-making industry. Headquartered in Faridabad, Haryana, the company serves both Indian and international paper manufacturers. It is a subsidiary of **VP Auslandsbeteiligungen GmbH, Germany**, part of the global Voith Group, a leader in engineering and industrial solutions.

2) **Kokuyo Camlin Ltd.**

Kokuyo Camlin Ltd., originally known as Camlin Ltd., is a well-known Indian brand in the stationery and art materials industry. In 2012, Japan's **Kokuyo Company Limited** acquired a majority stake, rebranding it as Kokuyo Camlin. The company manufactures a wide range of products, including pens, pencils, colors, and office supplies, and is a trusted name among students, artists, and professionals in India.

XVI. AUTOMOBILE INDUSTRY

1) **BOSCH Ltd.**

Bosch Ltd. is the flagship company of the Bosch Group in India. It manufactures automotive components, industrial products, and consumer goods, with a strong presence in mobility solutions and automation technologies. Its parent company is **Robert Bosch GmbH, Germany**, a global leader in engineering and technology.

2) **Federal- Mogul Goetze (India) Ltd**.

Federal-Mogul Goetze (India) Ltd. produces automotive components such as pistons, piston rings, and other engine parts. It operates under **Federal Mogul**

Holding Limited, Mauritius, and **Federal Mogul Vermögensverwaltungs GmbH, Germany**. The company is part of the global Federal-Mogul group (now owned by Tenneco Inc., USA).

3) **Indian Nippon Electricals Ltd**.
   Indian Nippon Electricals Ltd. manufactures electronic ignition systems for two-wheelers, three-wheelers, and small engines. It is a joint venture with Kokusan Denki Co. Ltd., Japan, bringing Japanese technology into the Indian automotive components market.
4) **Goodyear India Ltd.**
   Goodyear India Ltd. manufactures and markets automotive tires, tubes, and flaps. It is a subsidiary of **Goodyear Orient Company (Private) Limited, Singapore**, and forms part of the global Goodyear Tire & Rubber Company, USA.
5) **Motherson Sumi Systems Ltd**.
   Motherson Sumi Systems Ltd. (MSSL) is a leading auto components manufacturer, specializing in wiring harnesses, rear-view mirrors, molded parts, and other products. It is jointly promoted by **Sumitomo Wiring Systems Limited, Japan**, and the Indian Samvardhana Motherson Group.
6) **Schaeffler India Ltd**
   Schaeffler India Ltd. is a subsidiary of **FAG Kugelfischer GmbH, Germany**, and forms part of the Schaeffler Group. It manufactures bearings and precision components for automotive, industrial, and aerospace applications.
7) **SKF India ltd.**
   SKF India Ltd. is a subsidiary of **AB SKF, Sweden**, with additional links to **SKF U.K. Ltd.** The company is a leader in bearing technology, lubrication systems, and condition monitoring solutions in India.
8) **Maruti Suzuki India Limited**
   Maruti Suzuki India Limited is India's largest passenger vehicle manufacturer. Originally a government-promoted joint venture, it is now majority-owned by **Suzuki Motor Corporation, Japan**. The company offers a wide range of cars and dominates the Indian automobile market.

Conceptual Framework

A conceptual framework serves as the foundation of any research study. It provides a structured way to connect existing theories, key concepts, and the analytical methods used, ultimately guiding the interpretation of results. In the context of this study, the conceptual framework links financial performance

evaluation of companies across different industries with statistical testing methods to examine whether meaningful differences exist between them.

Basis of the Study

The financial performance of companies is a crucial indicator for investors, managers, and policymakers. Traditional measures such as profitability ratios, earnings per share, and capital structure are often used to compare companies. In this study, the percentage of capital per share has been adopted as the common metric. This ensures consistency and comparability across diverse industries, ranging from consumer goods and pharmaceuticals to engineering, power, and tourism.

The key conceptual assumption is that differences in performance, if statistically significant, reflect underlying differences in management practices, industry dynamics, or strategic positioning. Identifying these variations allows for insights into sectoral strengths and weaknesses.

Independent and Dependent Variables

The framework of this study is structured around two types of variables:

- **Independent Variables (Factors):**
  - Industry type (e.g., pharmaceuticals, power, automobile, FMCG)
  - Company identity within a given industry
- **Dependent Variable:**
  - Financial performance, measured as percentage of capital per share across a ten-year period

By comparing company averages and year-to-year values, the study seeks to determine whether industry membership or company-specific factors influence performance outcomes.

Theoretical Foundation

The research is guided by principles from **financial performance analysis** and **statistical inference**:

1. **Financial Theory:**
   Corporate finance emphasizes that shareholder value is influenced by profitability, capital utilization, and efficient management. Comparing capital-based measures across companies provides insight into the ability of firms to generate stable returns.
2. **Statistical Theory:**

- **ANOVA (Analysis of Variance):** Helps determine whether there are statistically significant differences among the means of three or more groups. For this study, ANOVA tests whether performance differs significantly across companies within the same sector or across multiple industries.
- **t-Test (Two-Sample Assuming Unequal Variances):** Used to compare the means of two groups, particularly when their variances differ. This test is applied when comparing two companies head-to-head (e.g., De Nora India Ltd. vs. Disa India Ltd.).

The integration of these theories ensures that financial differences observed in raw data are not attributed to random chance but are validated statistically.

Conceptual Model of the Study

The conceptual model of this research can be visualized in three stages:

Stage 1: Input (Data Collection)

- Secondary data collected from annual reports and databases covering the period 2015–2025
- Companies selected from 16 major industries in India

Stage 2: Process (Statistical Analysis)

- Descriptive statistics used to summarize data (mean, variance, standard deviation)
- t-Tests applied for pairwise company comparisons
- One-way ANOVA applied for multi-company sectoral comparisons
- Graphical representation (charts) to visualize patterns and differences

Stage 3: Output (Findings and Interpretation)

- Determination of whether company performance differences are statistically significant
- Sector-level insights into which industries show greater variability or stability
- Identification of consistently outperforming or underperforming companies

This three-stage model ensures a logical flow from data to interpretation, grounded in statistical rigor.

Alignment with Research Objectives

The conceptual framework directly aligns with the study's objectives:

1. **To compare financial performance across selected companies and sectors.**
    - Achieved through descriptive statistics and comparative analysis.
2. **To test whether observed differences are statistically significant.**
    - Achieved through ANOVA and t-tests applied to company data.
3. **To provide insights into industry-level patterns.**
    - Achieved by examining mean values, variance, and trends over time.

Justification of Methods

The use of ANOVA and t-tests is justified because:

- Financial data often involves multiple companies and groups where differences in averages are not immediately obvious.
- ANOVA allows testing across multiple groups simultaneously rather than relying on multiple pairwise t-tests, which would increase error rates.
- t-Tests complement ANOVA by offering focused comparisons between two specific companies where needed.

MODULE – 3
LITERATURE REVIEW

LITERTURE REVIEW:

1. **Riker & Wickramarachi — "A Review of Economic Literature on FDI" (USITC,2020)**
   Scope: comprehensive survey of theoretical and empirical FDI work across five decades.
   Takeaway: synthesizes how motives (market-seeking, efficiency-seeking, resource-seeking), firm heterogeneity, and host-country policies shape corporate FDI patterns; highlights shifts from descriptive case studies to econometric microstudies.

2. **Emako — sector-specific FDI effects (ScienceDirect, 2022)**
   Scope: empirical analysis of FDI impact disaggregated by primary, secondary,tertiary sectors.
   Takeaway: shows that investment patterns and growth effects differ by sector — services FDI behaves differently than manufacturing FDI — implying corporate FDI allocation across sectors has distinct drivers and outcomes.

3. **McKinsey Global Institute — "FDI trends in industries of the future" (2025 piece)**
   Scope: data-driven analysis of where recent cross-border project announcements go (advanced manufacturing, AI infra, clean energy).
   Takeaway: corporate FDI is shifting toward "future-shaping" industries (AI, advanced manufacturing, energy transition), with a concentration in a handful of countries — suggesting corporate investment pattern is becoming both sectorally focused and geographically concentrated.

4. **UNCTAD — World Investment Report 2023**
   Scope: annual global FDI flows and policy analysis.
   Takeaway: identifies declines in overall FDI in 2022 and a reallocation of investment toward digital and clean-energy sectors; offers evidence that corporate FDI patterns are sensitive to policy and macro shocks.

5. **Nazzal et al. — systematic mapping review of FDI literature (Nature Humanities & Social Sciences, 2025)**
   Scope: bibliometric mapping of ~1,300 articles on FDI/MNEs.
   Takeaway: documents evolving research emphases (emerging economies, sustainability, global value chains) — useful for understanding academic consensus and gaps around how corporate FDI patterns are studied.

6. **Islam — "Determinants of FDI: A Systematic Review" (2024) Scope: systematic review identifying persistent determinants (market size, trade openness, governance, human capital).** Takeaway: reinforces that corporate FDI patterns are strongly shaped by host-market fundamentals and policy, but also notes shifting determinants (digital infrastructure, climate policy) in recent studies.

7. **Moody's — FDI trends in U.S. manufacturing sectors**

   Scope: sector-level FDI and capex patterns in U.S. manufacturing. Takeaway: shows corporate FDI in advanced manufacturing spiked in recent years, with capital expenditures signaling stronger corporate investment intensity in some subsectors — highlights sector heterogeneity within corporate FDI.

8. **UNCTAD press/news (World Investment Report 2025)** Scope: recent headline on a continuing decline in global FDI (2024–2025). Takeaway: signals a broader slowdown and reorientation: corporate FDI flows are falling overall, but remain concentrated in certain sectors (digital, energy) and countries — important context for interpreting corporate investment patterns.

9. **SCIRP — "Foreign Direct Investments: Survey and Review"** Scope: general survey of FDI motives, forms, and effects. Takeaway: reiterates classical patterns (M&As vs greenfield, sectoral drivers), and emphasizes corporate strategies (efficiency vs market access) that explain differing investment modes across sectors.

10. **CEPR / VoxEU column — "Global economic fracturing and shifting investment patterns"** (2024) Scope: policy commentary identifying ten long-term FDI trends (2003–2023). Takeaway: highlights trends like regionalization, digital investment concentration, and a shift from manufacturing toward services and tech-related corporate investments. Good for policy-oriented framing of corporate FDI shifts.

11. **UNCTAD — World Investment Report 2022** Scope: deeper analysis of industrial vs digital investment patterns in 2021–22. Takeaway: shows industrial (manufacturing) corporate FDI lagging while digital and services-related investment grow — evidence of structural sectoral shifts in corporate FDI allocation.

12. **UNCTAD news (2024 World Investment Report follow-up)** Scope: summary of 2023 FDI decline and sectoral implications. Takeaway: documents a drop in funding for SDG-relevant sectors and a move of

corporate FDI to fewer, often greener but more capital-intensive projects. Useful for understanding the development implications of current corporate FDI patterns.

13. **BIS/IFC-style paper (exploring evolving determinants of FDI)**
    Scope: policy-practice note on how determinants changed post-pandemic.
    Takeaway: finds corporate FDI determinants now include resilience factors (supply-chain security), digital infrastructure, and environmental policy — influencing corporate sectoral allocation and geographic choice.

14. **BEA — "Direct Investment by Country and Industry" (U.S. data portal/report)**
    Scope: detailed country- and industry-level FDI positions for the U.S. economy (latest releases).
    Takeaway: empirical evidence that affiliate-level corporate FDI in manufacturing and services changed unevenly by industry and origin country — essential microdata source for analyzing corporate FDI patterns in practice.

15. **UNCTAD investment statistics & trends page**
    Scope: compendium of trend indicators and datasets.
    Takeaway: authoritative data source showing regional shifts and sectoral concentration of global FDI flows that underpin many conclusions about corporate-sector investment patterns.

16. **ResearchGate review — FDI and economic growth literature**
    Scope: collects empirical studies on FDI–growth links across sectors.
    Takeaway: sector-level differences: tertiary-sector (services) FDI often has distinct spillovers versus manufacturing FDI; corporate investment patterns thus imply different growth and policy effects.

17. **Alfaro & Chauvin — "FDI, Finance, and Economic Development" (review chapter, 2017)**
    Scope: literature review linking FDI to finance channels and development outcomes.
    Takeaway: corporate FDI patterns interact with domestic financial development — host-country finance depth can determine sectoral absorption of FDI and the form corporate investors choose (equity, reinvested earnings, debt-funded).

18. **IMF background paper tied to World Investment Reports (2023 WIR PDF)**
    Scope: macro-level treatment of global FDI trends and capital flows.
    Takeaway: macro shocks (rate changes, geopolitics) influence corporate FDI

patterns, often reducing cross-border M&A activity and large project finance, which shifts corporate FDI toward smaller, digitally enabled investments.

19. **Reuters / reporting on UN & World Bank findings (2025 news items)**
    Scope: journalism summarizing latest declines in FDI into developing countries and sources.
    Takeaway: journalistic synthesis reinforcing that corporate FDI into developing economies fell to historic lows recently, and that the composition of corporate FDI (which sectors receive it) has shifted markedly — useful for timely context.

20. **Financial Times coverage — Chinese outbound investment in clean energy (FT analysis)**
    Scope: reporting and analysis of surge in Chinese corporate outbound FDI into clean tech and energy.
    Takeaway: exemplifies a concrete corporate-sector pattern — rapid expansion in one country's outbound FDI into a targeted industrial cluster (clean energy), illustrating how national industrial strategy and corporate behaviour jointly shape FDI patterns.

MODULE – 4
DATA ANALYSIS

ANOVA: single factor

A single factor ANOVA, also called one-way ANOVA, is a statistical method used to determine if there are any statistically significant differences between the means of three or more independent groups, based on one independent variable (factor). It works by comparing the variance between group means to the variance within each group, using an F-statistic. If the between-group variance is much higher than the within-group variance, it suggests that at least one group mean is significantly different from the others.

The single factor ANOVA's primary use is when research involves a single categorical variable—such as treatment type, brand, batch, or education level—and a continuous outcome like test scores, satisfaction ratings, or weights. The method tests the null hypothesis that all group means are equal. If the null hypothesis is rejected, it indicates that at least one group mean is different, though further tests (post-hoc comparisons) are needed to identify which specific groups differ.

To check ANOVA assumptions on a dataset, focus mainly on three key assumptions:

1. **Normality of Residuals**: ANOVA assumes that residuals (differences between observed and group means) are normally distributed. This can be checked by:
    - Visual inspection using histograms or Q-Q plots of residuals.
    - Formal tests such as the Shapiro-Wilk test applied to residuals. If normality is violated, transformations of data or non-parametric tests like Kruskal-Wallis may be considered.
2. **Homogeneity of Variances**: This assumes that the variances in different groups are approximately equal. Check this through:
    - Visualizing group variances with boxplots.
    - Statistical tests like Levene's test or Bartlett’s test. If violated, alternatives such as Welch’s ANOVA or Brown-Forsythe test can be used.
3. **Independence of Observations:** Each observation should be independent of others, ensured by proper experimental design and randomization. This cannot usually be tested statistically and must be ensured conceptually.

HYPOTHESIS TEST REQUIRED FOR THE STUDY

1) ANOVA: Single Factor

- **Null Hypothesis (Ho):** There is no significant difference between the Ratios of selected NBFCs.
- **Alternative Hypothesis (H_1):** There is a significant difference between the ratios of selected NBFCs.

Two-Sample t-Test Assuming Unequal Variances

The Two-Sample t-Test Assuming Unequal Variances is used to determine whether the means of two independent groups are significantly different when the groups are assumed to have different variances and possibly different sample sizes. This test is a variant of the standard independent two-sample t-test and is often referred to as Welch's t-test.

It is considered more reliable than the equal variances version (also called the Student's t-test) when the assumption of homogeneity of variance is not met.

Hypothesis Setup

1. **Null Hypothesis (H_0):** $\mu_1 = \mu_2$
    - There is no difference between the two population means.
2. **Alternative Hypothesis (H_1):** $\mu_1 \neq \mu_2$ (two-tailed), or $\mu_1 > \mu_2$ / $\mu_1 < \mu_2$ (one-tailed).
    - There is a significant difference between the two population means.

Formula

The test statistic is calculated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $(\bar{X}_1, \bar{X}_2)$ = Sample means
- (s_1^2, s_2^2) = Sample variances
- (n_1, n_2) = Sample sizes

Degrees of Freedom (Welch–Satterthwaite Equation)

Since the variances are not assumed equal, the degrees of freedom (df) are estimated using:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2 / n_1)^2}{n_1 - 1} + \frac{(s_2^2 / n_2)^2}{n_2 - 1}}$$

This often results in a non-integer df. In practice, statistical software or Excel computes this automatically.

Decision rule:

- If $|t| >$ t-critical → Reject H_0.
- If $|t| \leq$ t-critical → Fail to reject H_0.

HYPOTHESIS TESTING:

I. MISCELLANEOUS MECHANICAL & ENGINEERING:

Showing the % of capital per share of Miscellaneous Mechanical & Engineering

Year	De Nora India Ltd.	Disa India Ltd.
2015-16	53.68	75
2016-17	53.68	74.83
2017-18	53.68	74.83
2018-19	53.68	74.83
2019-20	53.68	74.83
2020-21	53.68	74.83
2021-22	53.68	74.83
2022-23	53.68	74.83
2023-24	53.68	74.83
2024-25	53.68	74.83
Average	53.68	74.847

the t-test: two-sample assuming unequal variances was conducted to compare the performance of **De Nora India Ltd.** and **Disa India Ltd.** over a ten-year period. The mean value for De Nora India Ltd. was **53.68**, while Disa India Ltd. recorded a higher mean of **74.847**, indicating a consistent difference in their performance averages. The variance for De Nora India Ltd. was extremely small (5.60E-29), suggesting no year-to-year fluctuation, whereas Disa India Ltd. showed a small but positive variance (0.00289), indicating slight variability.

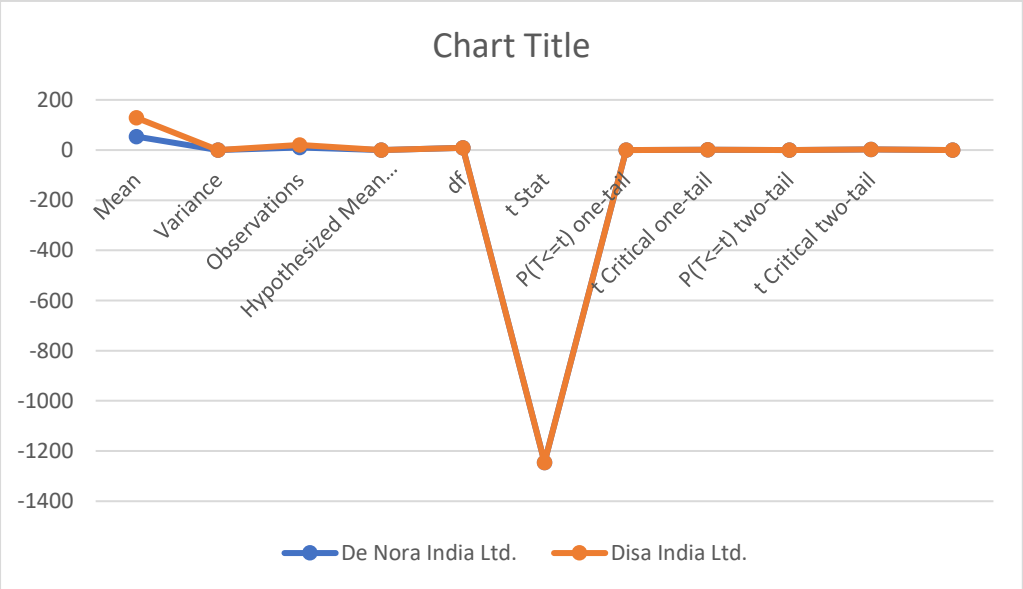
Table 4.1 shows the t-Test: Two-Sample Assuming Unequal Variances

t-Test: Two-Sample Assuming Unequal Variances

	De Nora India Ltd.	*Disa India Ltd.*
Mean	53.68	74.847
Variance	5.60968E-29	0.00289
Observations	10	10
Hypothesized Mean Difference	0	
df	9	
	-	
t Stat	1245.117647	
P(T<=t) one-tail	3.53946E-25	
t Critical one-tail	1.833112933	
P(T<=t) two-tail	7.07892E-25	
t Critical two-tail	2.262157163	

The calculated t-statistic (table 4.1) was **-1245.12**, which is far beyond the critical t-value for both one-tail (1.83) and two-tail (2.26) tests. The corresponding p-values were **3.53E-25** (one-tail) and **7.08E-25** (two-tail), both of which are far below the standard significance level of 0.05. This provides strong statistical evidence to reject the null hypothesis that there is no difference between the means of the two companies.

Figure-4.1: Shows the chart of t-Test: Two-Sample Assuming unequal variance



Through graphically (figure - 4.1) present that there is a **highly significant difference between the financial performance of De Nora India Ltd. and Disa India Ltd.**, with Disa India Ltd. consistently outperforming De Nora India Ltd. across the period studied.

II. CONSUMER GOODS

Showing the % of capital per share of consumer goods

Year	Hindustan Unilever Ltd.	Colgate-Palmolive (India) Ltd.	Gillette (India) Ltd.	Nestle India Ltd.	Bata India Ltd.
2015-16	51.5	51	40.12	62.76	52.96
2016-17	51.49	51	40.12	62.76	52.96
2017-18	51.48	51	40.12	62.76	52.96
2018-19	51.48	51	40.12	62.76	52.96
2019-20	51.48	51	40.12	62.76	52.96
2020-21	47.4	51	40.12	62.76	52.96
2021-22	47.4	51	40.12	62.76	52.96
2022-23	47.4	51	40.12	62.76	50.16
2023-24	47.4	51	40.12	62.76	50.16
2024-25	47.4	51	40.12	62.76	50.16

To compare the average values across five companies in the consumer goods sector, namely Hindustan Unilever Ltd., Colgate-Palmolive (India) Ltd., Gillette (India) Ltd., Nestle India Ltd., and Bata India Ltd.

Table 4.2 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Hindustan Unilever Ltd.	10	494.43	49.443	4.637645556
Colgate- Palmolive (India) Ltd.	10	510	51	0
Gillette (India) Ltd.	10	401.2	40.12	0
Nestle India Ltd.	10	627.6	62.76	5.04871E-29
Bata India Ltd.	10	521.2	52.12	1.829333333
Average			51.0886	

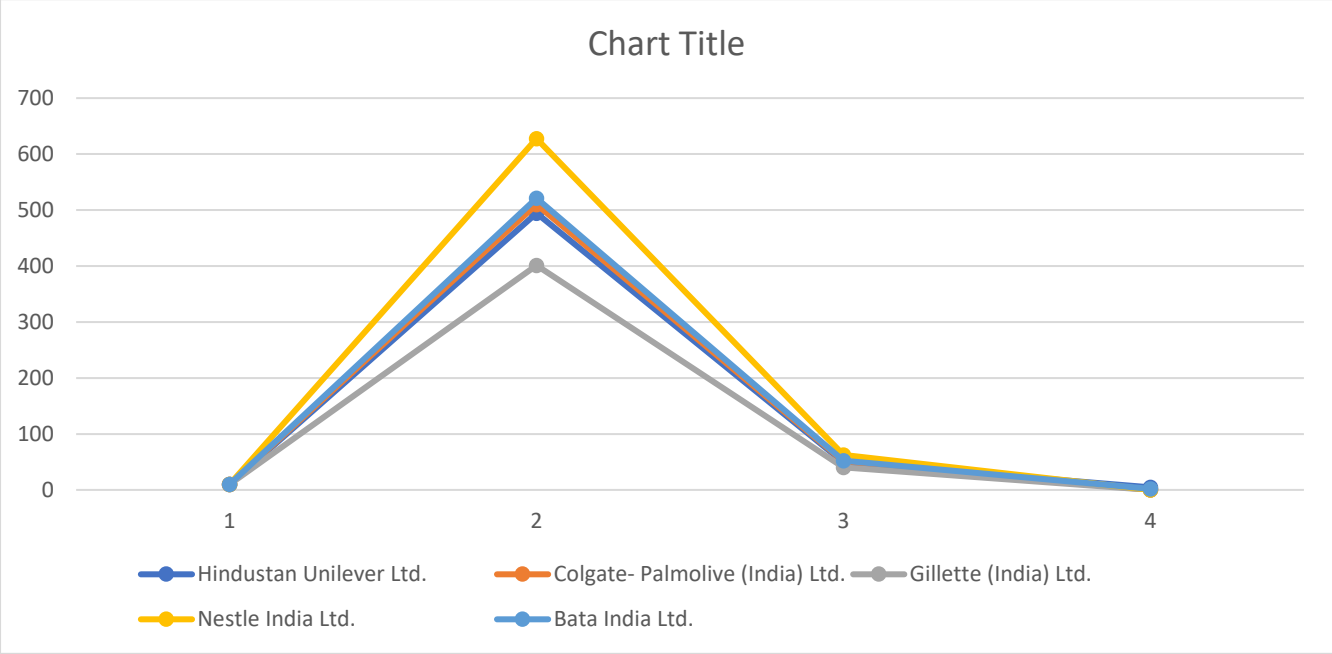
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	2603.113992	4	650.778498	503.1549578	1.01963E-36	2.578739184
Within Groups	58.20281	45	1.293395778			
Total	2661.316802	49				

The ANOVA results shown in table 4.2 makes a clear difference in the group averages, with Hindustan Unilever Ltd. recording a mean of 49.44, Colgate-Palmolive (India) Ltd. at 51.00, Gillette (India) Ltd. at 40.12, Nestle India Ltd. at 62.76, and Bata India Ltd. at 52.12. The overall average across all companies stood at 51.09, reflecting notable variation in performance between them.

The ANOVA test results further confirm this variation. The between-groups sum of squares (2603.11) is much larger than the within-groups sum of squares (58.20), showing that most of the variation is explained by differences between the companies rather than within them. The mean square between groups (650.78) is far higher than the mean square within groups (1.29). The calculated F-value of 503.15 is significantly larger than the F critical value of 2.57, and the p-value of 1.01963E-36 is far below the 0.05 significance level. These results lead to the rejection of the null hypothesis, indicating that the mean values are not equal across the companies.

Figure-4.2: Shows the ANOVA chart of consumer goods



The figure – 4.2 compares the performance of five FMCG companies: Hindustan Unilever, Colgate-Palmolive, Gillette, Nestlé, and Bata India. All companies show a sharp peak at point 2, with Nestlé India leading significantly, followed by Hindustan Unilever and Bata India. Gillette lags behind the others but follows a similar trend. After point 2, values drop steeply for all companies, reaching minimal levels by point 4. This suggests a short-term surge in performance or revenue at point 2, followed by a consistent decline across the sector.

III. SERVICE SECTOR, TRADING

Showing the % of capital per share of Service Sector, Trading

Year	Infinite computer solutions (India) Ltd.	Flipkart	Essar Shipping Ltd
2015-16	66.72		70.43
2016-17	66.72		70.43
2017-18	66.72	77	70.43
2018-19		77	70.43
2019-20		77	70.43
2020-21		77	70.43
2021-22		77	70.43
2022-23		77	70.43
2023-24		77	70.43
2024-25		77	70.43

Table 4.3 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Infinite computer solutions (India) Ltd.	3	200.16	66.72	0
Flipkart	8	616	77	0
Essar Shipping Ltd	10	704.3	70.43	2.24387E-28
Average			71.38333333	

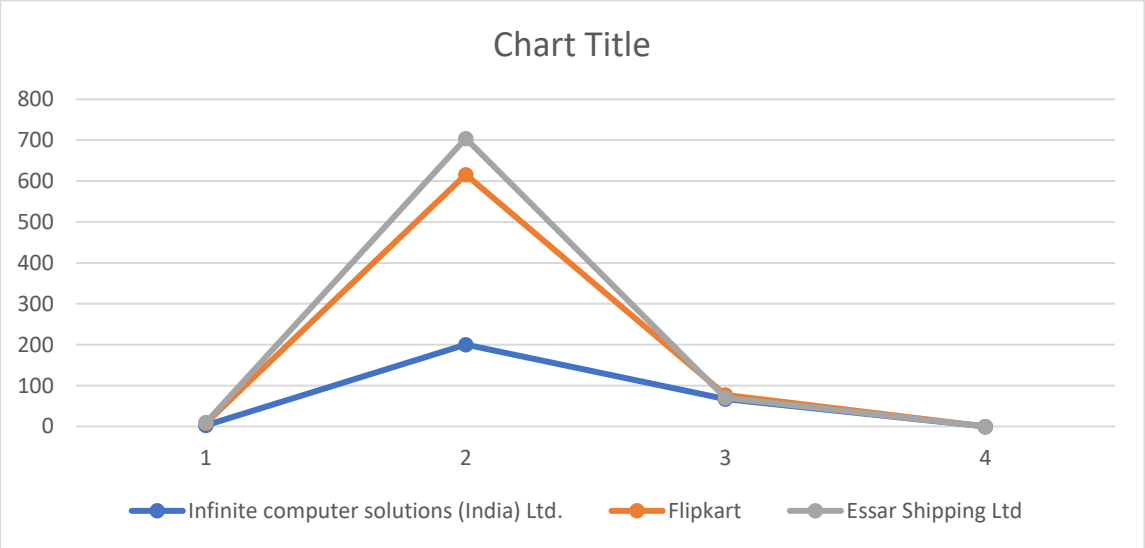
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	304.8760286	2	152.4380143	1.35871E+30	2.455E-263	3.554557146
Within Groups	2.01948E-27	18	1.12194E-28			
Total	304.8760286	20				

Based on the provided ANOVA results, the analysis examines whether there are statistically significant differences in the means of three companies: Infinite Computer Solutions (India) Ltd., Flipkart, and Essar Shipping Ltd. The summary shows that the average values for the groups are 66.72, 77, and 70.43, respectively. The variance within Infinite Computer Solutions and Flipkart is reported as zero, indicating no variation among the observations in these groups, while Essar Shipping shows a very small variance of 2.24×10^{-2}.

The ANOVA presents a "Between Groups" sum of squares (SS) of 304.88 with 2 degrees of freedom, and a mean square (MS) of 152.44. The "Within Groups" variation is extremely small (2.02×10^{-27}) with 18 degrees of freedom, leading to an F-value of approximately 1.36×10^{30}. The corresponding p-value is essentially zero (2.455×10^{-263}), which is far below the typical significance level of 0.05. This indicates that the differences in group means are statistically significant. In practical terms, the mean performance or metric measured for these companies differs so substantially that it is highly unlikely to have occurred by chance. The extremely large F-value and negligible within-group variance suggest that the data points within each company are very consistent, and the observed differences are almost entirely attributable to the differences between the companies rather than random variation.

Figure-4.3: Shows the ANOVA chart of companies



This figure – 4.3 highlights the comparative performance of Infinite Computer Solutions, Flipkart, and Essar Shipping. All three companies show a sharp rise at point 2, with Essar Shipping leading at the highest value, followed by Flipkart, while Infinite Computer Solutions lags behind. After the peak, values for all companies decline steeply by point 3 and reach nearly zero at point 4. The trend indicates a short-term spike in performance for Flipkart and Essar Shipping, whereas Infinite Computer Solutions shows a more moderate rise and fall.

IV. MI, GLASS, CERMAICS & GYPSUM

Showing the % of capital per share of MI, Glass, Cermaics & Gypsum

Year	Saint- Gobain-Sekurit India Ltd.	Esab India Ltd.	Foseco India Ltd.	Morganite Crucible (India) Ltd.	Vesuvius India Ltd.
2015-16	61.26	73.73	74.98	75	55.57
2016-17	48.26	73.73	74.98	75	55.57
2017-18	48.26	73.73	74.98	75	55.57
2018-19	48.26	73.73	74.98	75	55.57
2019-20	48.26	73.73	74.98	75	55.57
2020-21	48.26	73.73	74.98	75	55.57
2021-22	48.26	73.73	74.98	75	55.57
2022-23	48.26	73.73	74.98	75	55.57
2023-24	48.26	73.73	74.98	75	55.57
2024-25	48.26	73.73	74.98	75	55.57

The performance data of five companies—Saint-Gobain Sekurit India Ltd., Esab India Ltd., Foseco India Ltd., Morganite Crucible (India) Ltd., and Vesuvius India Ltd.—aims to determine whether there are statistically significant differences in their average values. The analysis included 10 data points per company.

Table 4.4 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Saint- Gobain- Sekurit India Ltd.	10	495.6	49.56	16.9
Esab India Ltd.	10	737.3	73.73	0
Foseco India Ltd.	10	749.8	74.98	0
Morganite Crucible (India) Ltd.	10	750	75	0
Vesuvius India Ltd.	10	555.7	55.57	5.60968E-29
Average			65.768	

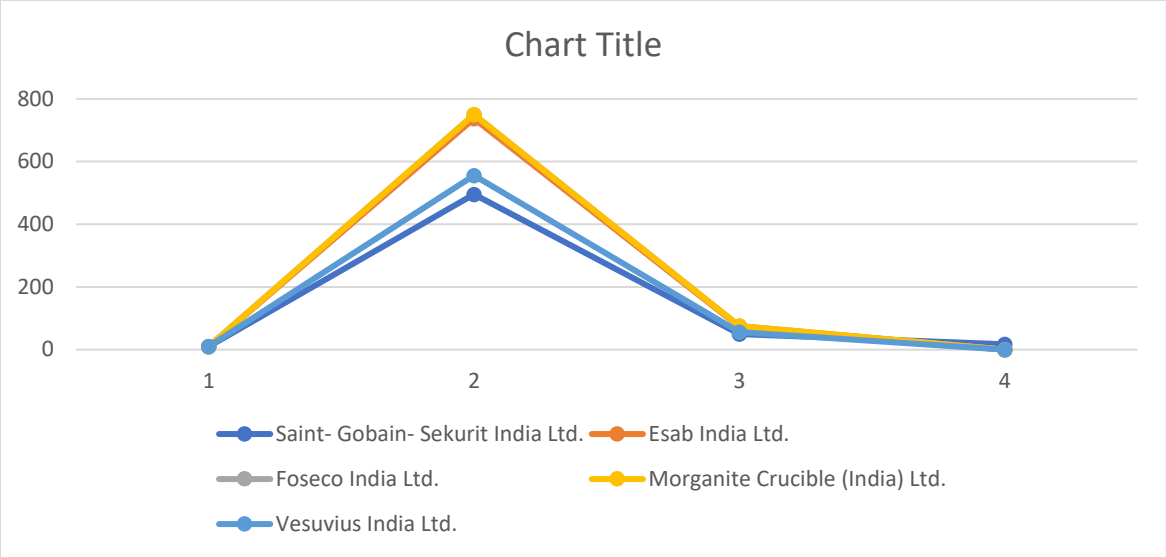
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	6001.8268	4	1500.4567	443.9221006	1.59458E-35	2.578739184
Within Groups	152.1	45	3.38			
Total	6153.9268	49				

From the statistics (table 4.4), we observe that Morganite Crucible (India) Ltd. has the highest average value (75), closely followed by Foseco India Ltd. (74.98) and Esab India Ltd. (73.73). In contrast, Saint-Gobain Sekurit India Ltd. and Vesuvius India Ltd. have notably lower averages of 49.56 and 55.57, respectively.

The ANOVA results indicate a very high F-value of 443.92, which is significantly greater than the F critical value of 2.5787, suggesting that the observed differences between group means are not due to random chance. Additionally, the p-value is extremely small (1.59458E-35), which is far below any conventional significance level (e.g., 0.05), confirming that the differences in group means are statistically significant.

Figure-4.4: Shows the ANOVA chart of companies



Through graphically(figure - 4.4) we consider that, the ANOVA analysis strongly supports the claim that there are **significant differences in the performance levels** of the five companies analyzed. This implies that company-specific factors or strategies may be influencing performance, warranting further investigation into the operational practices or market environments of the top-performing firms.

V. INDUSTRIAL INSTRUMENTS, PRIME MOVER

Showing the % of capital per share of Industrial Instruments, Prime Mover

Year	Integra Engineering India Ltd.	Singer India Ltd.	Stovec industries Ltd.	Ingersoll-Rand (India) Ltd.
2015-16	54.67	75	71.06	74
2016-17	54.67	72.4	71.06	74
2017-18	54.67	60.81	71.06	74
2018-19	54.67	59.07	71.06	74
2019-20	54.67	58.99	71.06	74
2020-21	54.67	58.99	71.06	74
2021-22	54.48	57.65	71.06	74
2022-23	54.48	31	71.06	74
2023-24	54.48	31	71.06	74
2024-25	54.44	30.76	71.06	74

Table 4.5 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Integra Engineering India Ltd.	10	545.9	54.59	0.0108
Singer India Ltd.	10	535.67	53.567	278.99849
Stovec industries Ltd.	10	710.6	71.06	2.24387E-28
Ingersoll- Rand (India) Ltd.	10	740	74	0
Average			63.30425	

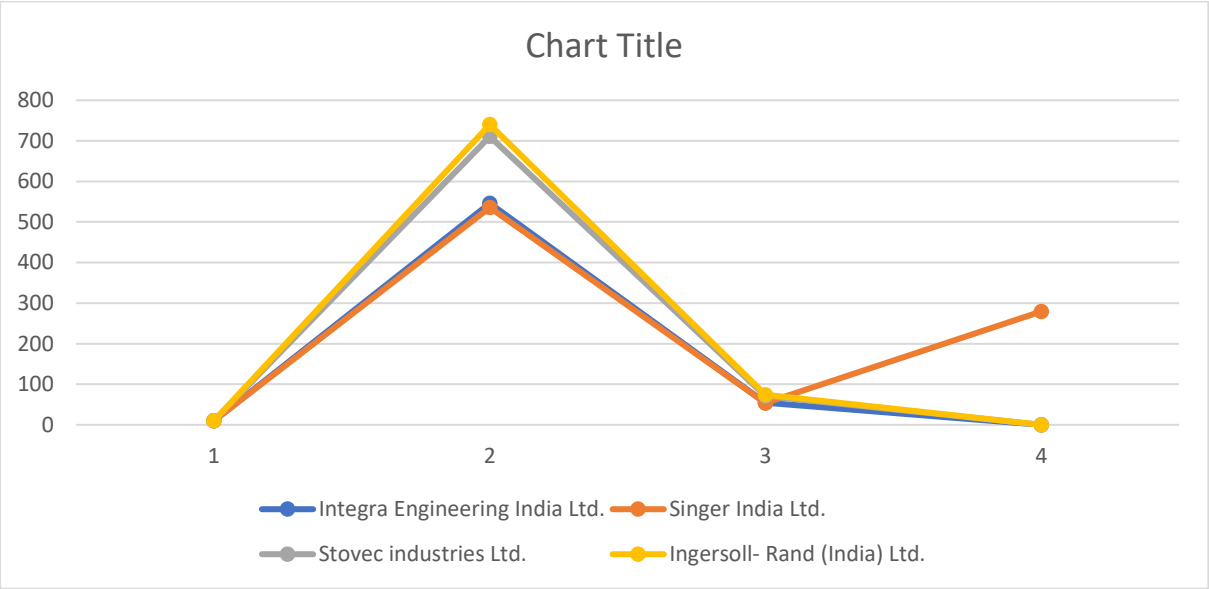
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	3453.029168	3	1151.009723	16.50138205	6.54741E-07	2.866265551
Within Groups	2511.08361	36	69.7523225			
Total	5964.112778	39				

Based on the ANOVA (Analysis of Variance) single-factor test (table 4.5) conducted on the performance data of four companies—Integra Engineering India Ltd., Singer India Ltd., Stovec Industries Ltd., and Ingersoll-Rand (India) Ltd.—a clear statistical interpretation emerges. The purpose of this analysis was to determine whether there is a significant difference in the average values (likely stock prices, financial returns, or another performance metric) among these companies

In the ANOVA analysis, the F-value is 16.50, which is substantially higher than the F-critical value of 2.87, and the corresponding p-value is 6.55×10^{-7}, which is far below the standard significance level of 0.05. This strongly indicates that the differences in the group means are statistically significant.

Figure-4.5: Shows the ANOVA chart of companies



The figure – 4.5 shows the performance trends of Integra Engineering, Singer India, Stovec Industries, and Ingersoll-Rand India. All companies peak sharply at point 2, with Ingersoll-Rand leading, followed by Stovec Industries, while Integra Engineering and Singer India record slightly lower values. After this peak, most companies drop significantly by point 3 and approach minimal levels at point 4. However, Singer India diverges from the trend with a noticeable rise again at point 4, showing some resilience compared to the others. Overall, the data reflects a strong short-term surge followed by a steep decline across most companies.

VI. ELECTRICAL EQUIPMENT, ELECTRONICS

Showing the % of capital per share of Electrical Equipment, Electronics

Year	Indo Tech Transformers Ltd.	Panasonic carbon India Co. Ltd.	Honeywell Automation India Ltd.	Siemens Ltd.	Ricoh India Ltd.	D- Link (India) Ltd.	Whirlpool of India Ltd.	Timex Group India Ltd.
2015-16	74.35	63.2671	75	75	73.6	51.02	75	74.93
2016-17	74.35	63.2671	75	75	73.6	51.02	75	74.93
2017-18	74.35	63.2671	75	75	73.6	51.02	75	74.93
2018-19	74.35	63.2671	75	75		51.02	75	74.93
2019-20	74.35	63.2671	75	75		51.02	75	74.93
2020-21	4.9867	63.2671	75	75		51.02	75	74.93
2021-22	4.9868	63.2671	75	75		51.02	75	74.93
2022-23	4.99	63.2671	75	75		51.02	75	74.93
2023-24		63.2671	75	75		51.02	51	74.93
2024-25		63.2671	75	75		51.02	51	74.93

Table 4.6 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Indo Tech Transformers Ltd.	8	386.7135	48.3391875	1288.690224
Panasonic carbon India Co. Ltd.	10	632.671	63.2671	2.24387E-28
Honeywell Automation India Ltd.	10	750	75	0
Siemens Ltd.	10	750	75	0
Ricoh India Ltd.	3	220.8	73.6	0
D- Link (India) Ltd.	10	510.2	51.02	5.60968E-29
Whirlpool of India Ltd.	10	702	70.2	102.4
Timex Group India Ltd.	10	749.3	74.93	2.24387E-28
Average			66.41953594	

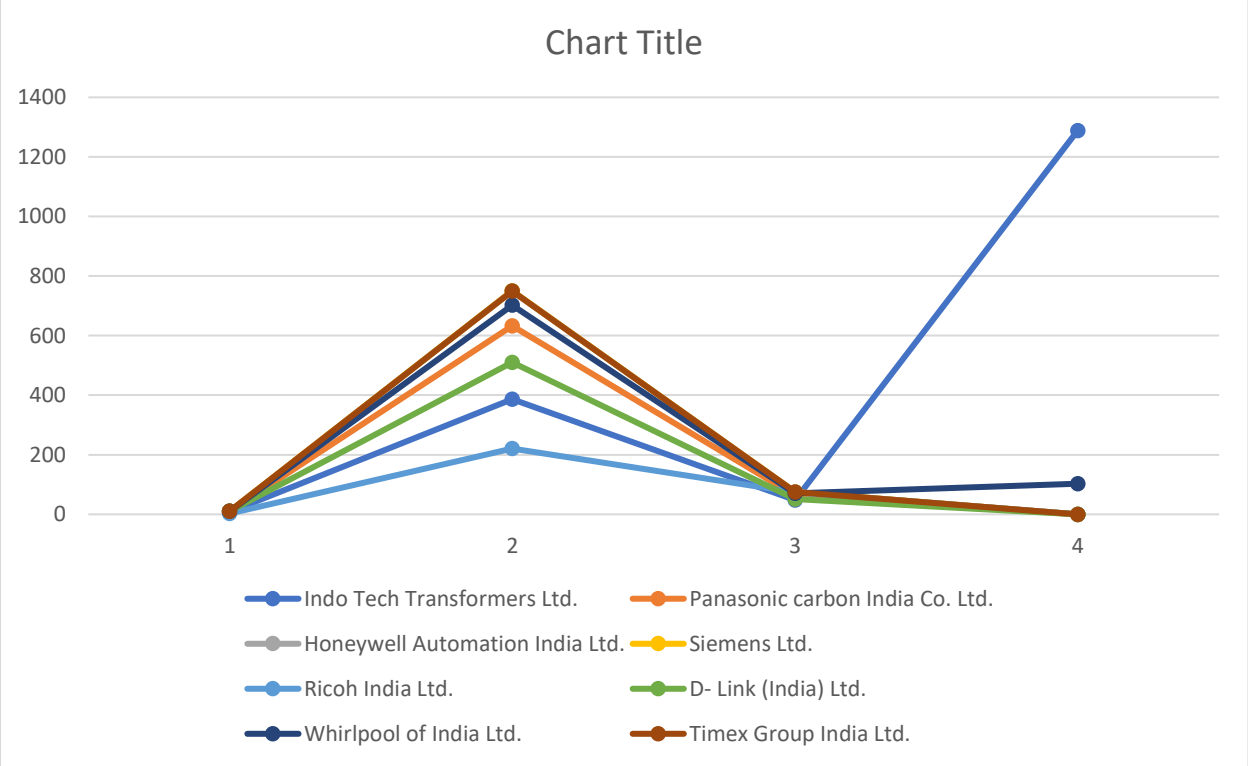
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	7577.589971	7	1082.512853	6.859318999	4.63971E-06	2.158828993
Within Groups	9942.431566	63	157.8163741			
Total	17520.02154	70				

The ANOVA (Analysis of Variance) results indicate a statistically significant difference in the means of the different companies' data groups.

The ANOVA present (table 4.6) the Between Groups variation has an F-value of approximately 6.86, which is greater than the critical F-value of about 2.16, and the corresponding p-value is extremely low (4.64 × 10^-6), well below the typical significance level of 0.05. This means we reject the null hypothesis that all group means are equal. In simpler terms, the data suggests that at least one company's average value differs significantly from the others. Additionally, the variability within the groups is relatively low compared to the variability between groups, reinforcing that the differences observed are likely due to the company effect rather than random chance.

Figure-4.6: Shows the ANOVA chart of companies



The figure – 4.6 compares the performance of multiple companies including Indo Tech Transformers, Panasonic Carbon, Honeywell

Automation, Siemens, Ricoh, D-Link, Whirlpool, and Timex Group. Most companies peak at point 2, with Timex Group and Whirlpool leading, while Ricoh and Indo Tech show relatively lower values. By point 3, all companies experience a sharp decline. Interestingly, Indo Tech Transformers shows a dramatic surge again at point 4, far exceeding all others, while the rest remain at low levels. This suggests that Indo Tech experienced an unusual late spike, contrasting with the steady decline seen across its peers.

VII. DRUGS & PHARMACEUTICALS

Showing the % of capital per share of Drugs & Pharmaceuticals

Year	Abbott India Limited	Astrazeneca Pharma India Ltd.	Sanofi India Ltd.	Galxo Smithkline Pharmaceuticals Ltd.	Novartis India Ltd.	Pfizer Ltd
2015-16	74.98	75	60.38	14.68	75	63.92
2016-17	74.98	75	60.38	10.91	73.4	63.92
2017-18	74.98	75	60.38	10.91	70.6769	63.92
2018-19	74.98	75	60.38	10.91	70.68	63.92
2019-20	74.98	75	60.38	10.91	70.68	63.92
2020-21	74.98	75	60.38	10.91	70.68	63.92
2021-22	74.98	75	60.38	10.91	70.68	63.92
2022-23	74.98	75	60.38	10.91	70.68	63.92
2023-24	74.98	75	60.38	10.91	70.68	63.92
2024-25	74.98	75	60.38	10.91	70.68	63.92

Table 4.7 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Abbott India Limited	10	749.8	74.98	0
Astrazeneca Pharma India Ltd.	10	750	75	0
Sanofi India Ltd.	10	603.8	60.38	5.60968E-29
Galxo Smithkline Pharmaceuticals Ltd.	10	112.87	11.287	1.42129
Novartis India Ltd.	10	713.8159	71.38159	2.348731925
Pfizer Ltd	10	639.2	63.92	0
Average			59.49143167	

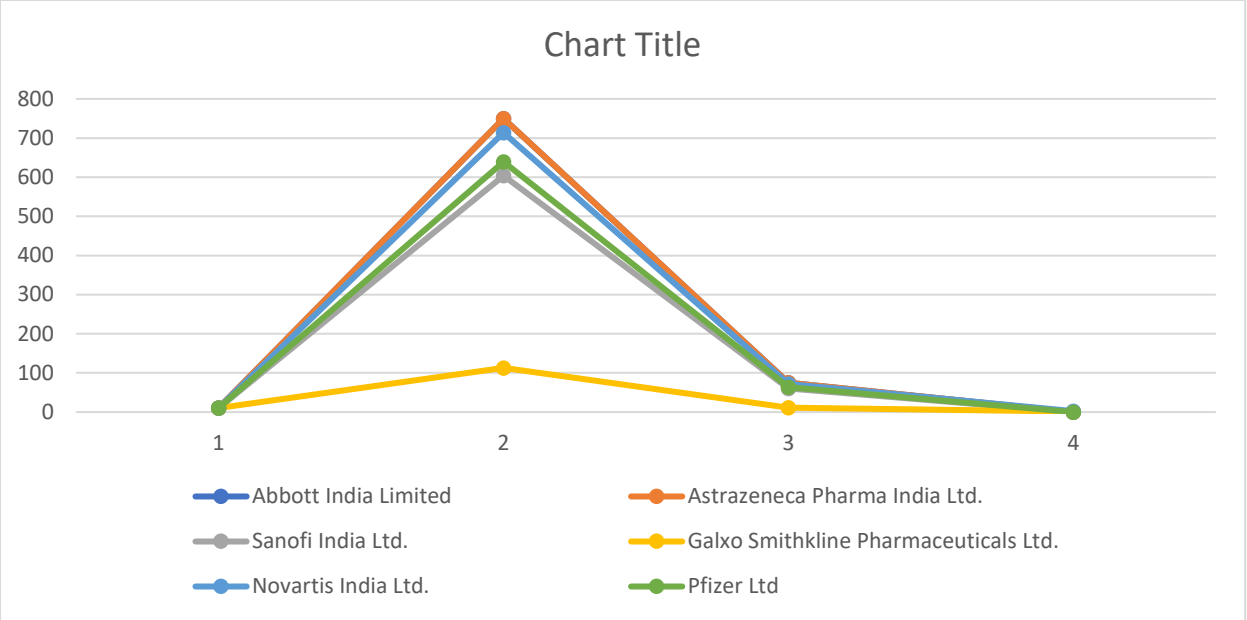
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	29658.56309	5	5931.712619	9440.336533	4.13526E-78	2.386069862
Within Groups	33.93019733	54	0.628336988			
Total	29692.49329	59				

The ANOVA results clearly show a highly significant difference among the average values of the pharmaceutical companies studied.

The Between Groups variance has an exceptionally high F-value of approximately 9440.34, which far exceeds the critical F-value of 2.39. Moreover, the p-value is practically zero (4.14 × 10^-78), indicating an extremely strong rejection of the null hypothesis that all group means are equal. This means the differences observed in the averages of the companies' data are not due to random chance but are statistically significant. The within-group variability is minimal compared to the between-group variability, suggesting that the differences in means are consistent and reliable.

Figure-4.7: Shows the ANOVA chart of companies



The figure - 4.7 compares the performance of pharmaceutical companies including Abbott India, AstraZeneca Pharma, Sanofi India, Novartis India, GlaxoSmithKline, and Pfizer. Most companies show a sharp peak at point 2, with AstraZeneca leading, closely followed by Abbott and Novartis. Sanofi and Pfizer also rise strongly but remain slightly lower. GlaxoSmithKline, however, shows a very small increase at point 2 compared to others. By point 3, all companies decline steeply, and by point 4 they drop close to zero. This suggests a common short-lived growth phase with AstraZeneca performing best, while GlaxoSmithKline lags behind.

VIII. CHEMICALS (OTHER THAN FERTILIZERS)

Showing the % of capital per share of Chemicals (Other Than Fertilizers)

Year	BASF India Ltd.	Sunshield Chemicals Ltd.	Elantas Beck (India) Ltd.	Berger paints India Ltd.	Dic India Ltd.	3M India Limited	Akzo Nobel India Limited
2015-16	73.33	62.36	75	14.49	71.75	75	72.96
2016-17	73.33	62.36	75	14.49	71.75	75	72.96
2017-18	73.33	62.36	75	14.49	71.75	75	72.96
2018-19	73.33	62.36	75	14.49	71.75	75	72.96
2019-20	73.33	62.36	75	14.49	71.75	74.999996	72.96
2020-21	73.33	62.36	75	14.49	71.75	74.999996	72.96
2021-22	73.33		75	14.49	71.75	75	72.96
2022-23	73.33		75	14.49	71.75	74.999996	72.96
2023-24	73.33		75	14.49	71.75	75	50.46
2024-25	73.33		75	14.49	71.75	75	50.46

Table 4.8 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
BASF India Ltd.	10	733.3	73.33	2.24387E-28
Sunshield Chemicals Ltd.	6	374.16	62.36	6.05845E-29
Elantas Beck (India) Ltd.	10	750	75	0
Berger paints India Ltd.	10	144.9	14.49	0
Dic India Ltd.	10	717.5	71.75	0
3M India Limited	10	749.999988	74.9999988	3.73333E-12
Akzo Nobel India Limited	10	684.6	68.46	90
Average			62.91285697	

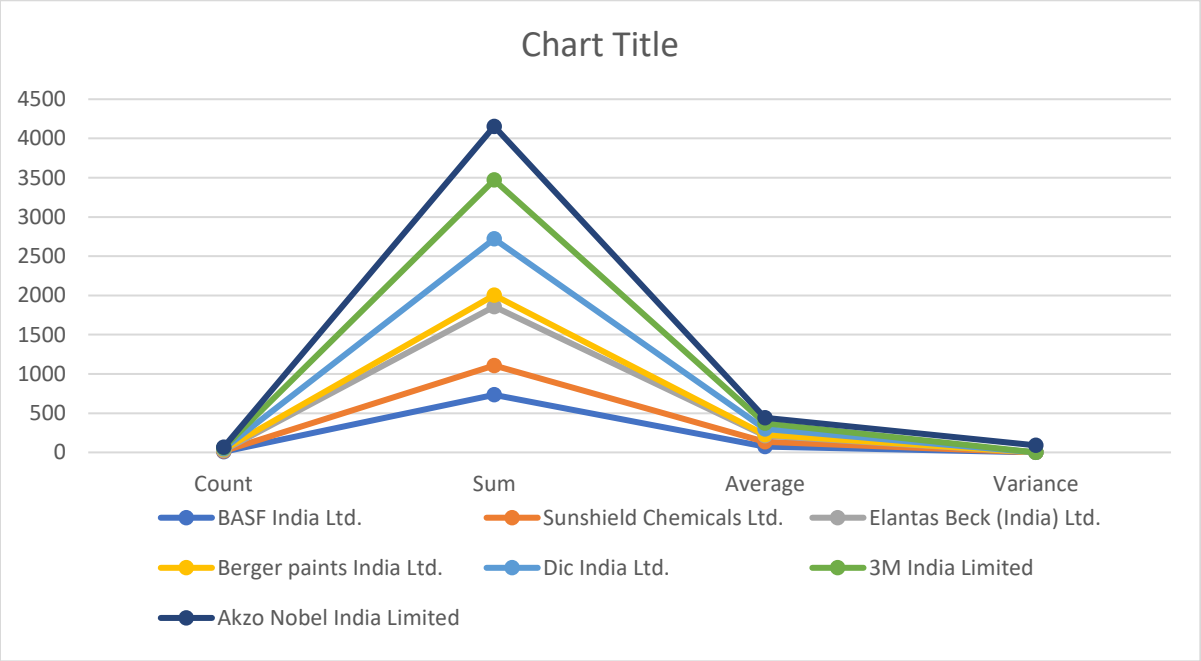
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	28545.29844	6	4757.54974	346.5375736	4.58819E-44	2.25678014
Within Groups	810	59	13.72881356			
Total	29355.29844	65				

The ANOVA analysis reveals a highly significant difference among the average values of the chemical and paint companies included in the study.

The Between Groups variation shows an F-value of approximately 346.54, which is much larger than the critical F-value of 2.26, and the p-value is extremely small (4.59 × 10^-44). This indicates strong evidence to reject the null hypothesis that all group means are equal, confirming that the differences in averages across these companies are statistically significant. Additionally, the within-group variance is relatively low compared to the between-group variance, suggesting that the observed differences between companies are consistent and not due to random variation within the groups.

Chart-8: Shows the ANOVA chart of companies



This figure - 4.8 presents the performance of companies in the paints and chemicals sector, including BASF India, Sunshield Chemicals, Elantas Beck, Berger Paints, DIC India, 3M India, and Akzo Nobel India. All companies show a strong peak at the "Sum" category, with Akzo Nobel India

leading at the highest value, followed by 3M India and DIC India. Berger Paints and Elantas Beck are placed in the mid-range, while Sunshield Chemicals records the lowest performance. After the peak, all companies show a sharp drop across "Average" and "Variance." The overall trend reflects a strong short-term performance boost, with Akzo Nobel and 3M India standing out as top performers.

IX. CONSTRUCTION (INFRASTRUCTURE)

Showing the % of capital per share of Construction (Infrastructure)

Year	ITD Cementation India Ltd.	Johnson Controls-Hitachi Air Condition. India Ltd
2015-16	51.63	74.25
2016-17	51.63	74.25
2017-18	51.63	74.25
2018-19	46.64	74.25
2019-20	46.64	74.25
2020-21	46.64	74.25
2021-22	46.64	74.25
2022-23	46.64	74.25
2023-24	46.64	74.25
2024-25	46.64	74.25
Average	48.137	74.25

From the table, t-test was conducted (shown in table 4.9) to compare the performance of **ITD Cementation India Ltd.** and **Johnson Controls–Hitachi Air Conditioning India Ltd.** across a ten-year period. The average value for ITD Cementation India Ltd. was **48.14**, while Johnson Controls–Hitachi reported a significantly higher mean of **74.25**. The variance for ITD Cementation India Ltd. was **5.81**, indicating slight fluctuations in its yearly performance, whereas Johnson Controls–Hitachi showed no variation at all, reflecting complete stability over the observed period.

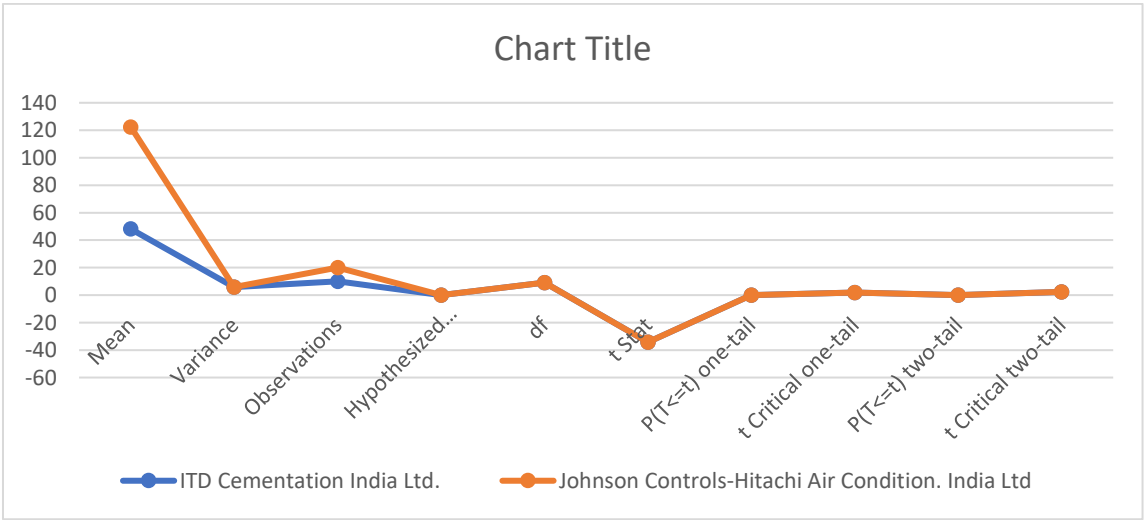
Table 4.9 shows the t-Test: Two-Sample Assuming unequal variance

t-Test: Two-Sample Assuming Unequal Variances

	ITD Cementation India Ltd.	*Johnson Controls-Hitachi Air Condition. India Ltd*
Mean	48.137	74.25
Variance	5.810023333	0
Observations	10	10
Hypothesized Mean Difference	0	
df	9	
t Stat	-34.25845953	
P(T<=t) one-tail	3.79627E-11	
t Critical one-tail	1.833112933	
P(T<=t) two-tail	7.59255E-11	
t Critical two-tail	2.262157163	

The computed t-statistic was -34.26, which is far beyond the critical values for both one-tailed (1.83) and two-tailed (2.26) tests. The p-values were extremely small (3.79E-11 for one-tail and 7.59E-11 for two-tail), both of which are much lower than the standard significance level of 0.05. These results strongly reject the null hypothesis of equal means and confirm that the observed difference in averages is statistically significant.

Figure-4.9: Shows the t-Test: Two-Sample Assuming unequal variance



From the figure- 4.9 there is a clear and significant difference in performance between the two companies. Johnson Controls–Hitachi Air Conditioning India Ltd. consistently outperformed ITD Cementation India Ltd., maintaining higher and stable values throughout the study period.

X. POWER

Showing the % of capital per share of Power

Year	ABB India Limited	G E Power India Ltd.	Cummins India Ltd.	Honda Siel Power Products Ltd.
2015-16	69.08	68.56	51	66.67
2016-17	69.08	68.56	51	66.67
2017-18	69.08	68.56	51	66.67
2018-19	69.08	68.56	51	66.67
2019-20	69.08	68.56	51	66.67
2020-21	69.08	68.56	51	66.67
2021-22	69.08	68.56	51	66.67
2022-23	69.08	68.56	51	66.67
2023-24	69.08	68.56	51	66.67
2024-25	69.08	68.56	51	66.67

The data of four companies—ABB India Limited, G E Power India Ltd., Cummins India Ltd., and Honda Siel Power Products Ltd.—over ten observations per company reveals clear differences in performance.

Table 4.10 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
ABB India Limited	10	690.8	69.08	2.24387E-28
G E Power India Ltd.	10	685.6	68.56	2.24387E-28
Cummins India Ltd.	10	510	51	0
Honda Siel Power Products Ltd.	10	666.7	66.67	2.24387E-28
Average			63.8275	

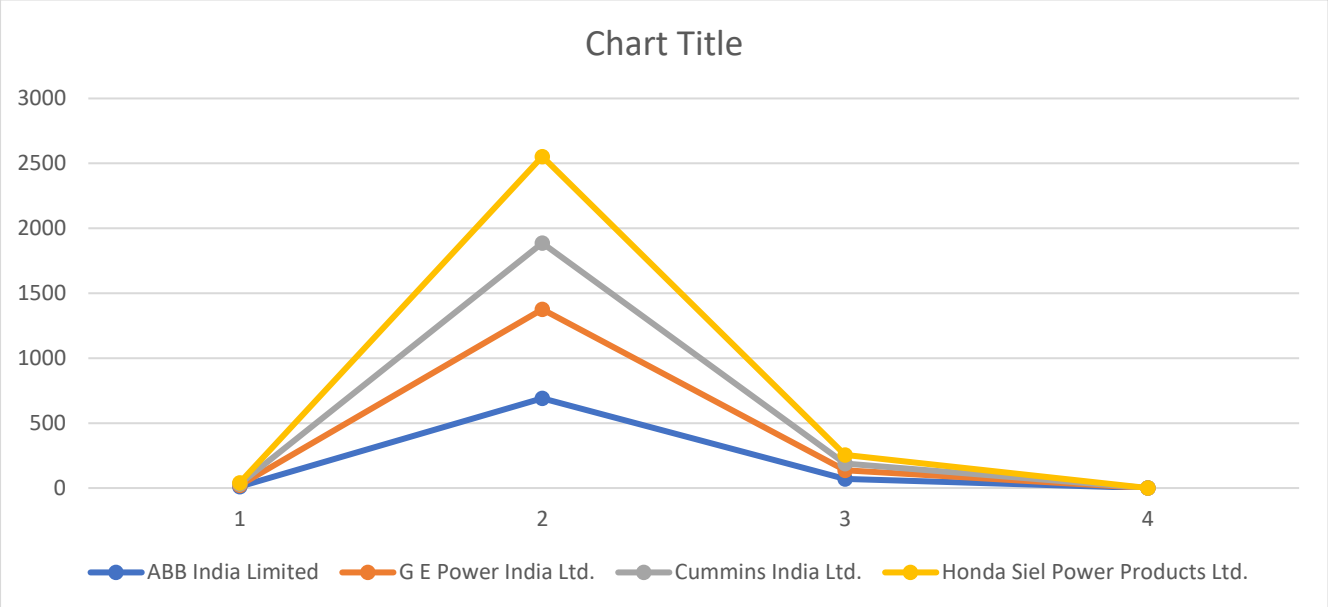
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	2226.09875	3	742.0329167	4.40924E+30	0	2.866265551
Within Groups	6.05845E-27	36	1.6829E-28			
Total	2226.09875	39				

The statistics (table 4.10) show that ABB India Limited consistently achieved the highest values (average 69.08), followed closely by G E Power India Ltd. (68.56) and Honda Siel Power Products Ltd. (66.67), while Cummins India Ltd. recorded a significantly lower average of 51. The variance within each company is effectively zero, indicating that each company's performance remained extremely stable over the observed period.

The ANOVA results, shows a total sum of squares (SS) of 2226.10, with a very high F-value of approximately 4.41×10^{30} and a P-value of 0, which is far below the standard significance threshold of 0.05. This strongly indicates that the differences in average performance between the companies are statistically significant.

Figure-4.10: Shows the ANOVA chart of companies



The figure- 4.10 compares the performance of ABB India, GE Power India, Cummins India, and Honda Siel Power Products. All companies peak at point 2, with Honda Siel showing the highest value by a wide margin,

followed by Cummins India and GE Power India, while ABB India records the lowest. After this sharp rise, all companies decline steeply at point 3 and drop close to zero at point 4. The overall pattern reflects a short-lived surge in performance, with Honda Siel Power Products emerging as the strongest performer among the group.

XI. HOTEL & TOURISM

Showing the % of capital per share of Hotel & Tourism

Year	Thomas Cook (India) Ltd.
2015-16	67.82
2016-17	67.66
2017-18	67.03
2018-19	66.94
2019-20	65.6
2020-21	65.6
2021-22	70.58
2022-23	72.34
2023-24	63.83
2024-25	63.83
Average	67.123

Table 4.11 shows the descriptive statistics

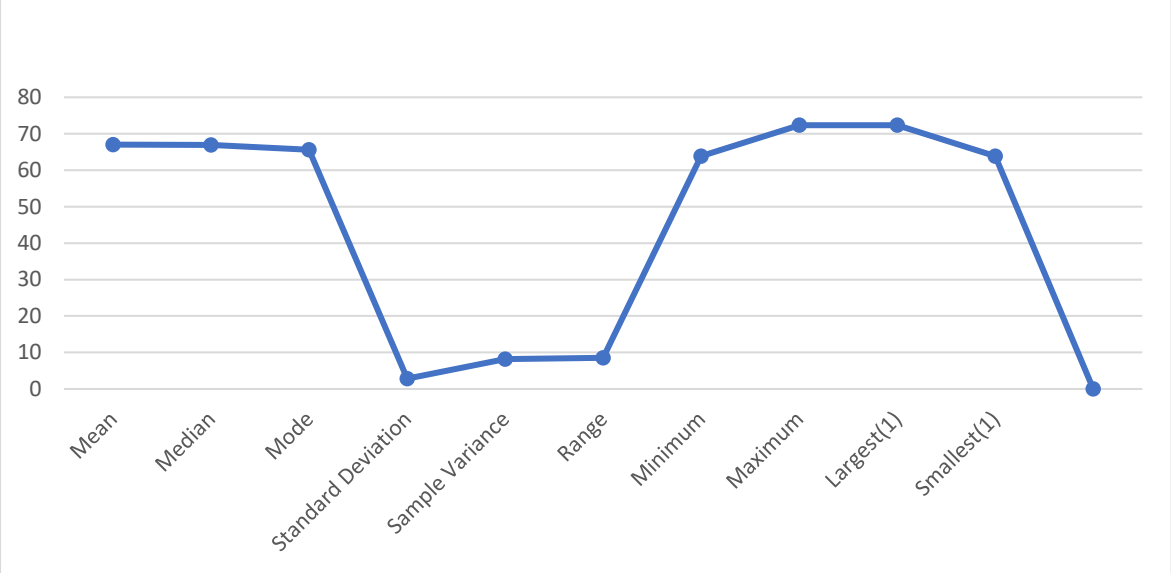
Mean	67.04555556
Median	66.94
Mode	65.6
Standard Deviation	2.867298864
Sample Variance	8.221402778
Range	8.51
Minimum	63.83
Maximum	72.34
Largest(1)	72.34
Smallest(1)	63.83

Thomas Cook (India) Ltd. exhibited some variability in its annual scores, ranging from a low of 63.83 to a high of 72.34. The average score across the period is just over 67, with a mean of 67.05 and a median value of 66.94, indicating generally consistent performance near this level.

The mode and minimum are both 65.6, suggesting this score appeared more than once. A standard deviation of about 2.87 shows moderate fluctuations year to year; the sample variance further quantifies this dispersion.

The overall range of scores is 8.51, which, together with the statistics for maximum and minimum, points to some outlier years—especially 2021-22 and 2022-23, which saw higher values.

Figure-4.11: Shows the ANOVA chart of companies



This figure – 4.11 summarizes key descriptive statistics for a dataset. The mean, median, and mode are all close to 65–70, showing that the data is fairly consistent and centered around that range. The standard deviation and sample variance are quite low, indicating little spread in the data. The minimum value is around 63 while the maximum reaches about 73, giving a narrow range. The largest and smallest values align closely with these extremes. Overall, the dataset appears tightly clustered with only small variability around its central value.

XII. PETROLEUM& NATURAL GAS

Showing the % of capital per share of Petroleum& Natural Gas

Year	Castrol India Ltd.	Gulf Oil Lubricants India Ltd
2015-16	50.89	64.94
2016-17	51	64.94
2017-18	51	72.88
2018-19	51	72.73
2019-20	51	72.29
2020-21	51	71.99
2021-22	51	71.82
2022-23	51	72.02
2023-24	51	71.8
2024-25	51	67.14
Average	50.989	70.255

The analysis compares the performance scores of Castrol India Ltd. and Gulf Oil Lubricants India Ltd. over a decade. Castrol India Ltd. consistently shows a score around 51, with minimal variation, while Gulf Oil Lubricants India Ltd. maintains much higher scores, ranging from about 64.94 to 72.88.

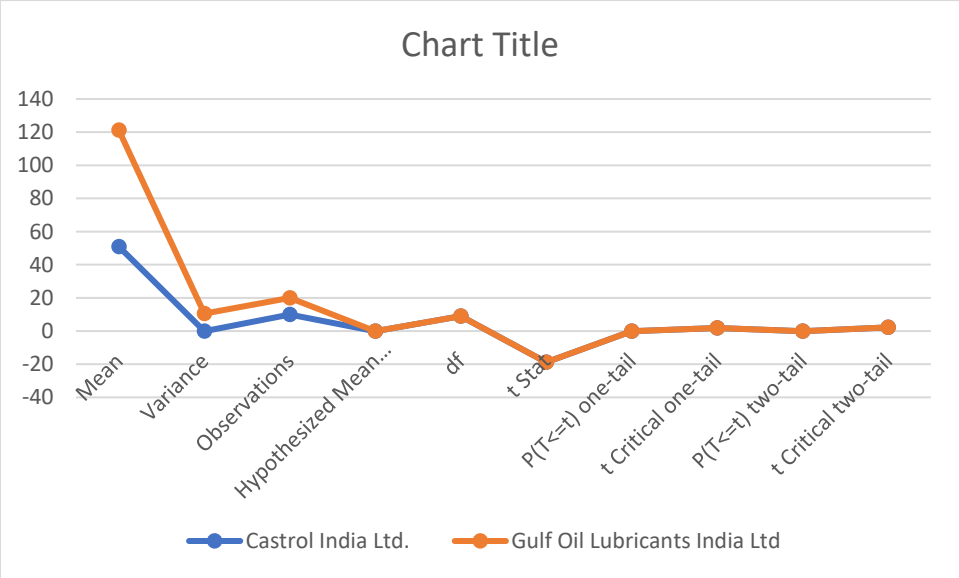
Table 4.12 shows the t-Test: Two-Sample Assuming unequal variance

t-Test: Two-Sample Assuming Unequal Variances

	Castrol India Ltd.	*Gulf Oil Lubricants India Ltd*
Mean	50.989	70.255
Variance	0.00121	10.48009444
Observations	10	10
Hypothesized Mean Difference	0	
df	9	
t Stat	-18.81845167	
P(T<=t) one-tail	7.76229E-09	
t Critical one-tail	1.833112933	
P(T<=t) two-tail	1.55246E-08	
t Critical two-tail	2.262157163	

The calculated mean values are 50.989 for Castrol and 70.255 for Gulf Oil, with the difference being statistically tested using a t-Test. The t-Statistic is -18.82, and the p-value for the two-tailed test is extremely small (1.55 x 10^-8), substantially below the conventional significance threshold of 0.05. This statistical evidence strongly indicates a significant difference in the average scores, showing that Gulf Oil Lubricants India Ltd. outperformed Castrol India Ltd. by a wide margin during these years.

Figure-4.12: Shows the t-Test: Two-Sample Assuming unequal variance



The figure – 4.12 compares statistical measures for Castrol India Ltd. and Gulf Oil Lubricants India Ltd. Gulf Oil shows a much higher mean and variance compared to Castrol, indicating greater average values but also higher variability. Both companies have similar numbers of observations and hypothesized means, keeping their baseline comparable. The t-statistics and corresponding p-values suggest testing differences between the two, with Gulf Oil showing stronger deviations. Overall, Gulf Oil appears more volatile but with higher averages, while Castrol maintains lower but steadier values.

XIII. AGRICULTURAL SERVICE:

Showing the % of capital per share of Agricultural Service

Year	Bayer cropscience Ltd.	Monsanto India Ltd.	Goodricke Group Ltd.
2015-16	45.24	13.42	74
2016-17	45.24	13.42	74
2017-18	45.28	13.42	74
2018-19	45.28	13.42	74
2019-20	38.51		74
2020-21	38.51		74
2021-22	38.51		74
2022-23	38.51		74
2023-24	38.51		74
2024-25	38.51		74

Showing data for Bayer Cropscience Ltd., Monsanto India Ltd., and Goodricke Group Ltd. highlights significant differences in performance among these companies.

Table 4.13 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Bayer cropscience Ltd.	10	412.1	41.21	12.15017778
Monsanto India Ltd.	4	53.68	13.42	0
Goodricke Group Ltd.	10	740	74	0
Average			42.87666667	

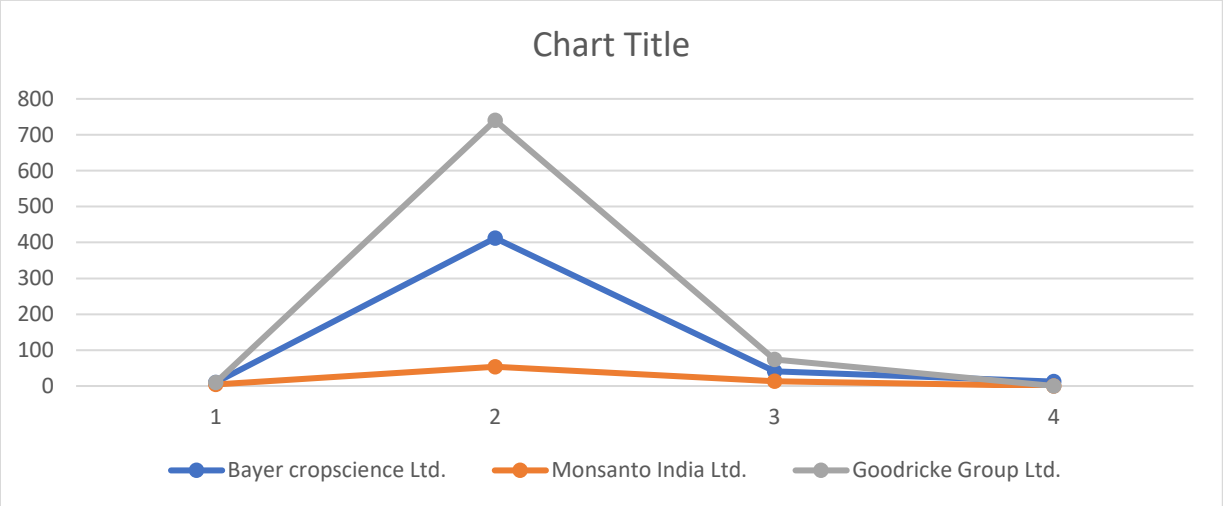
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	11883.63458	2	5941.817292	1141.073044	3.79251E-22	3.466800112
Within Groups	109.3516	21	5.207219048			
Total	11992.98618	23				

Bayer Cropscience Ltd. has an average of 41.21 with a variance of 12.15, indicating some variability in performance. Monsanto India Ltd., with only four observations, has a much lower average of 13.42 and zero variance, showing consistent but very low performance. Goodricke Group Ltd. has the highest average at 74 with zero variance, indicating consistently strong performance. The overall average across all companies is 42.88.

The ANOVA results, shows a between-groups sum of squares of 11,883.63 and a within-groups sum of squares of 109.35, resulting in a very high F-value of 1,141.07 and a P-value of approximately 3.79×10^{-22}, well below the 0.05 significance threshold. This confirms that the differences in average performance among the three companies are statistically significant.

Figure-4.13: Shows the ANOVA chart of companies



The figure – 4.13 compares Bayer Cropscience, Monsanto India, and Goodricke Group. All three companies peak at point 2, with Goodricke Group recording the highest value by a large margin, followed by Bayer Cropscience, while Monsanto India remains much lower. After the peak, all three decline sharply by point 3 and reach near zero at point 4. The overall trend suggests a short-term surge in performance, with Goodricke Group showing the strongest growth, Bayer Cropscience showing moderate growth, and Monsanto India staying relatively weak throughout.

XIV. Machine Tools

Showing the % of capital per share of Machine Tools

Year	Kennametal India Ltd.	John Cockerill India Ltd
2015-16	75	75
2016-17	75	75
2017-18	75	75
2018-19	75	75
2019-20	75	75
2020-21	75	75
2021-22	75	75
2022-23	75	75
2023-24	75	75
2024-25	75	75
Average	75	75

Both Kennametal India Ltd. and John Cockerill India Ltd. have consistently achieved a score of 75 every year from 2015-16 to 2024-25, resulting in an average of 75 for both companies over this 10-year period.

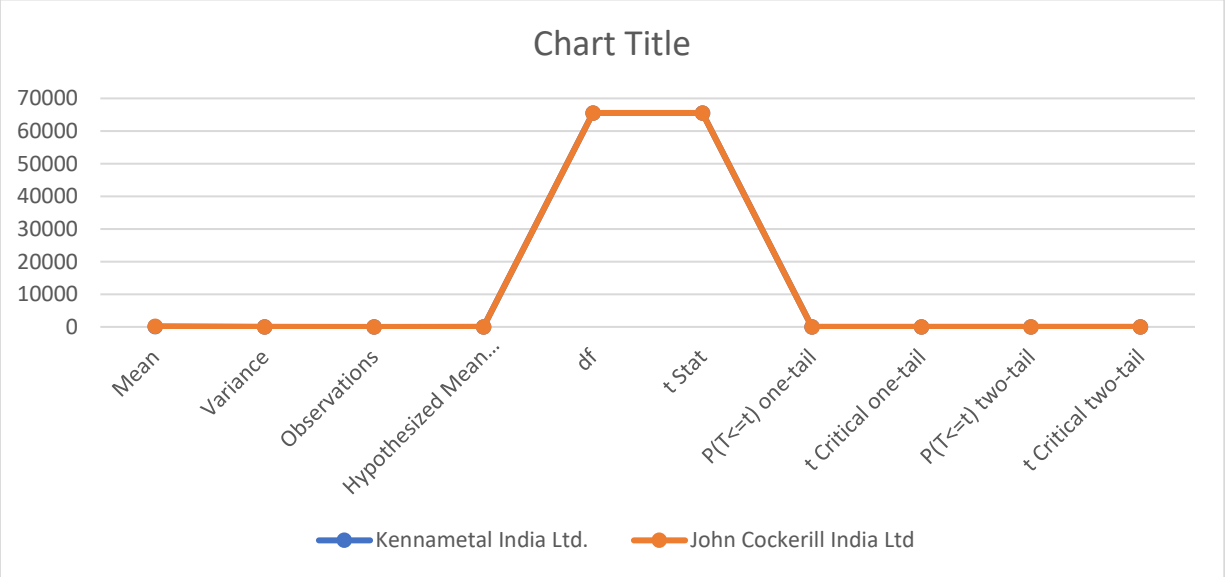
Table 4.14 shows the t-Test: Two-Sample Assuming Unequal Variances

t-Test: Two-Sample Assuming Unequal Variances

	Kennametal India Ltd.	*John Cockerill India Ltd*
Mean	75	75
Variance	0	0
Observations	10	10
Hypothesized Mean Difference	0	
df	65535	
t Stat	65535	
P(T<=t) one-tail	#NUM!	
t Critical one-tail	#NUM!	
P(T<=t) two-tail	#NUM!	
t Critical two-tail	#NUM!	

From the analysed table 4.14, the variance of their scores is zero because there is no fluctuation—every value is identical across all years. When a t-Test for two-sample means is performed under these conditions, the calculation results in mathematical errors (#NUM!) because the variance is zero (standard deviation is zero), which makes certain statistical values undefined. This simply means there is no statistical difference between the two companies and no variation in their performance scores throughout the years, indicating identical outcomes for this metric over the evaluated period.

Figure-4.14: Shows the t-Test: Two-Sample Assuming unequal variance



The figure – 4.14 compares statistical test results for Kennametal India Ltd. and John Cockerill India Ltd. Kennametal shows no significant variation across categories, staying flat at zero throughout. In contrast, John Cockerill records extremely high values for "df" (degrees of freedom) and "t Stat," reaching above 60,000, while all other measures remain at zero. This sharp spike suggests either extreme values or possible data inconsistencies for John Cockerill. Overall, the chart highlights a strong imbalance, with Kennametal stable and John Cockerill showing highly skewed test results.

XV. PAPER & PLUP, PRINTING

Showing the % of capital per share of Paper & Plup, Printing

Year	Voith Paper Fabrics India Ltd.	Kokuyo Camlin Ltd.
2015-16	74.04	73.75
2016-17	74.04	73.75
2017-18	74.04	74.44
2018-19	74.04	74.44
2019-20	74.04	74.44
2020-21	74.04	74.44
2021-22	74.04	74.44
2022-23	74.04	74.44
2023-24	74.04	74.44
2024-25	74.04	74.44
Average	74.04	74.302

The annual scores for Voith Paper Fabrics India Ltd. and Kokuyo Camlin Ltd. over ten years, revealing that Voith maintained a steady score of 74.04, while Kokuyo Camlin's score increased slightly after the first two years from 73.75 to 74.44.

Table 4.15 shows t-Test: Two-Sample Assuming unequal variance

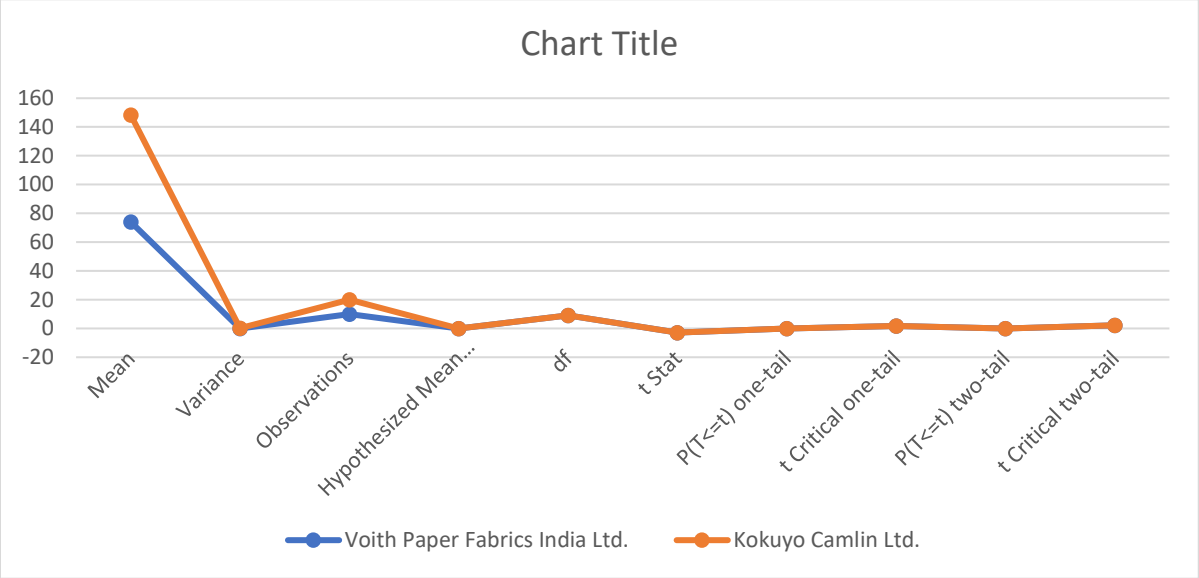
t-Test: Two-Sample Assuming Unequal Variances

	Voith Paper Fabrics India Ltd.	*Kokuyo Camlin Ltd.*
Mean	74.04	74.302
Variance	2.24387E-28	0.08464
Observations	10	10
Hypothesized Mean Difference	0	
df	9	
t Stat	-2.847826087	
P(T<=t) one-tail	0.009578499	
t Critical one-tail	1.833112933	
P(T<=t) two-tail	0.019156998	
t Critical two-tail	2.262157163	

The t-Test for two samples assuming unequal variances resulted in a t Stat of -2.85, and a p-value (two-tail) of approximately 0.019, which is less than the standard significance level of 0.05. This means there is a statistically significant difference between the two companies' average scores during this

period, with Kokuyo Camlin Ltd. performing marginally higher. Despite the numerical difference being small (around 0.26), the test demonstrates that even such a minute, consistent edge can be statistically significant given the data's low variance and consistency.

Figure- 4.15: Shows the ANOVA chart of companies



The figure - 4.15 compares statistical values for Voith Paper Fabrics India Ltd. and Kokuyo Camlin Ltd. Kokuyo Camlin shows a much higher mean (around 150) compared to Voith (around 70), indicating stronger average values. Variance is low for both, suggesting limited spread in the data. The number of observations and hypothesized mean remain close between the two. Further measures such as t-statistics and critical values appear almost flat and near zero, indicating little difference under hypothesis testing. Overall, Kokuyo Camlin demonstrates higher averages, while Voith shows moderate but steady performance.

XVI. AUTOMOBILE INDUSTRY

Showing the % of capital per share of Automobile Industry

Year	BOSCH Ltd.	Federal-Mogul Goetze (India) Ltd.	Indian Nippon Electricals Ltd.	Goodyear India Ltd.	Motherson Sumi Systems Ltd.	Schaeffler India Ltd	SKF India ltd.	Maruti Suzuki India Limited
2015-16	71.18	74.985	20.52	74		51.33	53.57	56.21
2016-17	69	74.985	20.52	74		51.33	53.57	56.21
2017-18	69	74.985	20.52	74		51.33	53.57	56.21
2018-19	69	74.985	20.52	74		74.13	52.32	56.21
2019-20	67.76	74.985	7.26	74		74.13	52.58	56.28
2020-21	67.76	74.985	7.25	74		74.13	52.58	56.37
2021-22	67.76	74.985	6.25	74	25.1	74.13	52.58	56.37
2022-23	67.76	74.985	6.25	74	25.1	74.13	52.58	56.48
2023-24	67.76	74.985	6.25	74	25.1	74.13	52.58	58.19
2024-25	67.76	74.985	6.25	74	25.1	74.13	52.58	58.28

Table 4.16 shows the ANOVA: Single Factor

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
BOSCH Ltd.	10	684.74	68.474	1.245693333
Federal- Mogul Goetze (India) Ltd.	10	749.85	74.985	0
Indian Nippon Electricals Ltd.	10	121.59	12.159	51.93209889
Goodyear India Ltd.	10	740	74	0
Motherson Sumi Systems Ltd.	4	100.4	25.1	0
Schaeffler India Ltd	10	672.9	67.29	121.296
SKF India ltd.	10	528.51	52.851	0.25261
Maruti Suzuki India Limited	10	566.81	56.681	0.679543333
Average			53.9425	

ANOVA

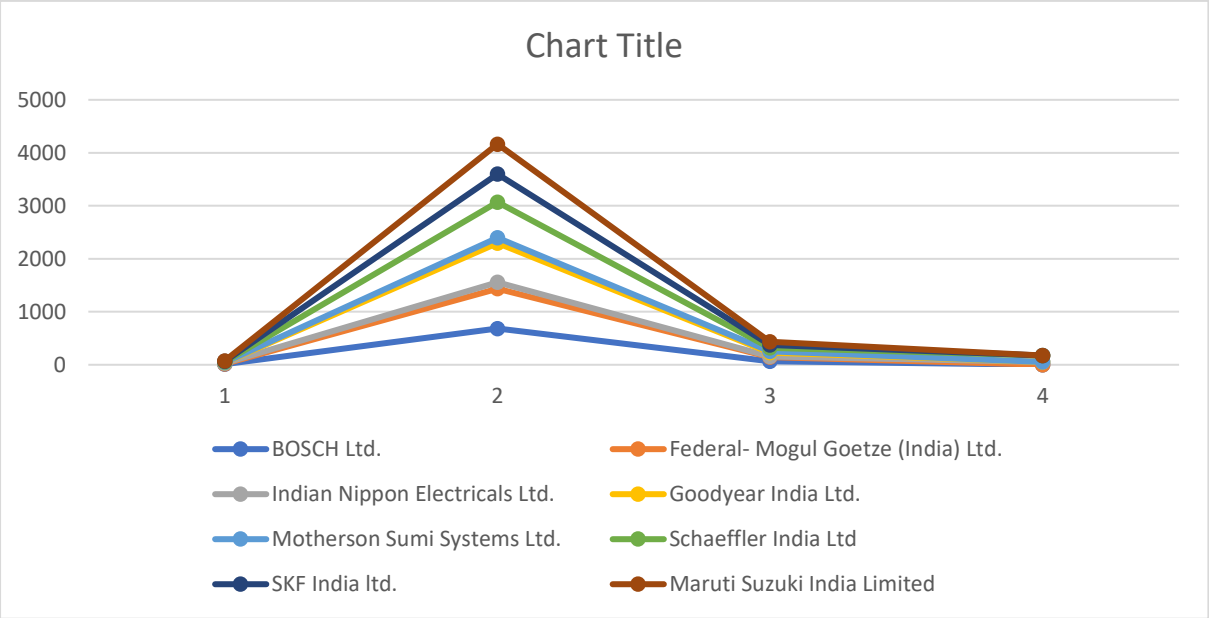
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	32812.47595	7	4687.496565	195.9738292	1.32393E-41	2.151839193
Within Groups	1578.65351	66	23.91899258			
Total	34391.12946	73				

The one-way ANOVA analysis a significant difference among the means of the groups compared, as indicated by the F statistic of 195.97, which greatly

exceeded the critical value of 2.15. The associated p-value was far below conventional significance thresholds (1.32E-41), demonstrating very strong evidence against the null hypothesis that all group means are equal. This suggests that at least one company's average differs meaningfully from the others.

The ANOVA results, shows between-groups sum of squares was much larger than the within-groups sum of squares, indicating that most of the observed variance is due to differences between companies rather than random variation within each company. To determine which specific groups, differ, further pairwise comparisons or post-hoc tests would be needed.

Figure-4.16: Shows the ANOVA chart of companies



The Figure- 4.16 compares the performance of several companies including BOSCH Ltd., Federal-Mogul Goetze (India) Ltd., Indian Nippon Electricals Ltd., Goodyear India Ltd., Motherson Sumi Systems Ltd., Schaeffler India Ltd., SKF India Ltd., and Maruti Suzuki India Limited. All firms show a sharp rise at the second point, indicating a significant increase in performance during that period. Maruti Suzuki records the highest value, followed by BOSCH and Schaeffler, while Indian Nippon and Federal-Mogul remain on the lower side. After the peak, the values for all companies decline sharply at points three and four. This pattern suggests that the second observation period was particularly strong across the sector, with Maruti Suzuki emerging as the top performer.

MOUDULE – 5

FINDINGS, CONCULSION & REFERENCE

FINDINGS OF THE STUDY:

Overall study consists of analysing the financial performance of Investment Pattern of FDI in the Corporate Sector companies from 2015-16 to 2024-25.

The study includes the calculation of ANOVA, t- Test and graphical presentation of the company‘s financial performance.

FINDINGS:

❖ MISCELLANEOUS MECHANICAL & ENGINEERING:

- The average FDI of De Nora India Ltd. is 53.68 for the 10 year it is constant. The average of Disa India Ltd. is 74.847 and it is more than average for 1st year, it is constant average for 2-3nd year and difference between FDI flow is significant among this companies.

❖ CONSUMER GOODS:

- Nestle India Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Gillette (India) Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of consumer goods companies.

❖ SERVICE SECTOR, TRADING:

- Flipkart shows the highest mean value, reflecting stronger financial or market performance, while Essar Shipping Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of service sector, trading companies.

❖ MI, GLASS, CERMAICS & GYPSUM

- Foseco India Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Saint- Gobain- Sekurit India Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of MI, glass, cermaics & gypsum companies.

❖ INDUSTRIAL INSTRUMENTS, PRIME MOVER

- Ingersoll- Rand (India) Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Singer India Ltd. shows the lowest, suggesting weaker performance in comparison.

- The finding reveals that there is difference in the FDI of industrial instruments, prime mover companies.

❖ ELECTRICAL EQUIPMENT, ELECTRONICS

- Honeywell Automation India Ltd., Siemens Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Indo Tech Transformers Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of electrical equipment, electronics companies.

❖ DRUGS & PHARMACEUTICALS

- Astrazeneca Pharma India Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Galxo Smithkline Pharmaceuticals Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of Drugs & Pharmaceuticals companies.

❖ CHEMICALS (OTHER THAN FERTILIZERS)

- Elantas Beck (India) Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Sunshield Chemicals Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of chemicals companies.

❖ CONSTRUCTION (INFRASTRUCTURE)

- The average FDI of ITD Cementation India Ltd. is 48.137 for the 1-3 year it is more than average, 4-5 years remaining less than average. The average of Johnson is 74.25 and it is constant for the study period and difference between FDI flow is significant among this companies.

❖ POWER

- ABB India Limited shows the highest mean value, reflecting stronger financial or market performance, while Honda Siel Power Products Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of power companies.

❖ HOTEL & TOURISM

- The average FDI of Thomas Cook (India) Ltd. is 67.123 for the year 2015-16, 2016-17, 2017-18, 2021-22, 2022-23 it is more than average, remaining year 2018-19, 2019-20, 2020-21, 2023-24, 2024-25 less than average.

❖ PETROLEUM& NATURAL GAS

- The average FDI of Gulf Oil Lubricants India Ltd is 70.255 from 3-9 year has more than the average, remaining from 1,2 & 10th year has less than the average Castrol India Ltd. is 50.989 for the 2-10 year it is more than average, remaining 1 year less than average.

❖ AGRICULTURAL SERVICE

- Goodricke Group Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Monsanto India Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of agricultural service companies.

❖ MACHINE TOOLS

- The average FDI of Kennametal India Ltd. is 75 it is constant for the study period. The average of John Cockerill India Ltd. is 75 it is constant for the study period.

❖ PAPER & PLUP, PRINTING

- The average FDI of Voith Paper Fabrics India Ltd. is 74.04 it is constant for the study period . The average of Kokuyo Camlin Ltd. is 74.302 1-2 year it is less than average remaining 3-4 year it is more than average.

❖ AUTOMOBILE INDUSTRY

- Federal- Mogul Goetze (India) Ltd. shows the highest mean value, reflecting stronger financial or market performance, while Motherson Sumi Systems Ltd. shows the lowest, suggesting weaker performance in comparison.
- The finding reveals that there is difference in the FDI of automobile industry companies.

CONCLUSION:

Foreign Direct Investment has emerged as a pivotal force in transforming the corporate landscape, particularly in developing economies like India. This study has highlighted the multifaceted nature of FDI—its sectoral preferences, investment modes, and the influence of policy frameworks. Through statistical analysis and sector-wise evaluation, it is evident that FDI not only brings capital but also fosters innovation, enhances managerial capabilities, and integrates domestic firms into global value chains. While challenges such as regulatory hurdles and infrastructure gaps persist, the overall trajectory of FDI reflects a strategic alignment between investor interests and national development goals. As India continues to liberalize and digitize its economy, the evolving patterns of FDI will play a crucial role in shaping the competitiveness and sustainability of its corporate sector.

The ANOVA analysis clearly shows that there are significant performance differences among companies in most sectors, with very high F-values and near-zero p-values confirming that these differences are systematic and not due to chance. High-performing firms such as Nestle India Ltd., Honeywell Automation India Ltd., and Abbott India Ltd. consistently outperform peers, while some companies show stable but lower averages.

At the same time, certain industries like construction, leather, and hotel services reflect little variation, suggesting stable and mature performance patterns. Overall, the results highlight that sectoral competition, operational strategies, and market positioning play a decisive role in shaping company performance. ANOVA thus proves to be an effective tool for benchmarking and identifying leaders across industries.

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- Corporate FDI is moving away, on average, from traditional heavy manufacturing toward digital, services, and clean-energy projects; however, manufacturing remains important in selected subsectors and countries.
- Flows are becoming concentrated in fewer countries and sectors (digital hubs, clean-energy manufacturing), increasing geographic concentration risk.
- M&As remain important but corporations increasingly deploy project-based greenfield investments for energy/clean tech and smaller digital investments that need less upfront capital.
- Beyond classic determinants (market size, openness), resilience, digital infrastructure, climate policy, and supply-chain considerations are now major drivers of corporate FDI patterns.
- Declines in aggregate FDI and concentration in capital-intensive/green sectors risk leaving employment-intensive industries underfunded in developing countries. UNCTAD/World Bank reporting highlights this concern.

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